

tastepocket

the pocket-conditioned finetuning set, instance by instance

379 T1 chemosensory complexes from the PDB become 225 (ligand, receptor) records, of which 201 decompose. Every record carries a 24-bit flavour vector and a 1280-d ESM-2 embedding of the 10 Å pocket around its ligand. Insect chemoreception is excluded.

What is in it

corpus records	201
decompositions	610 (3.03 per record)
R-groups	741 (1.21 per decomposition)
distinct ligands (CCD)	130
distinct receptors (UniProt)	73
receptor families	14
organisms	16
convec width	1304 = 24 flavour + 1280 pocket
pocket residues (mean)	67.0
pocket sites behind them	1086

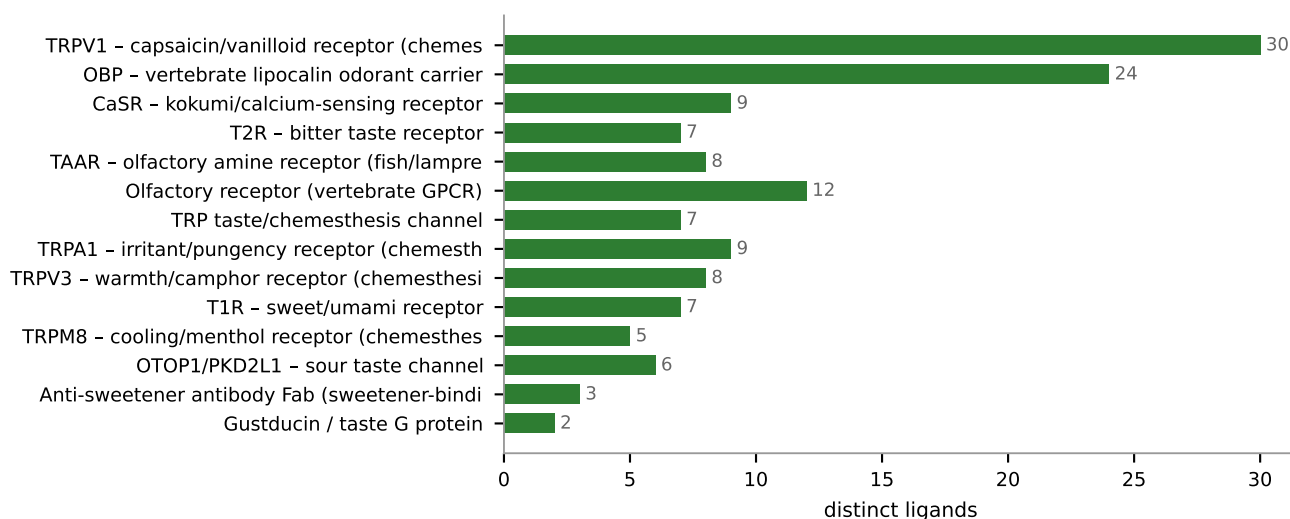
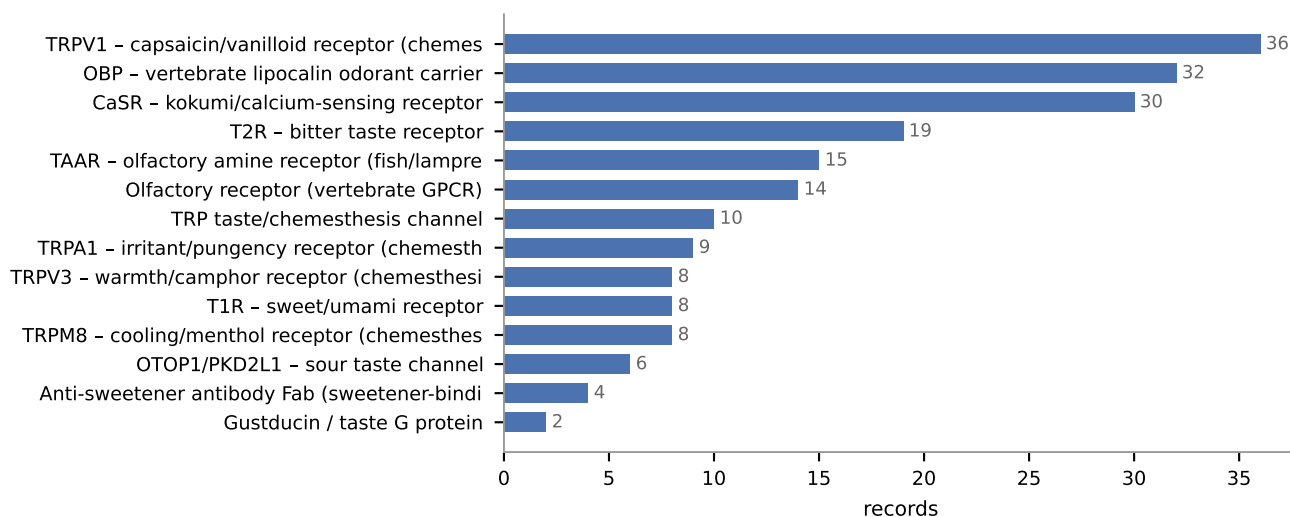
Why the counts are what they are

One record per (ligand, RECEPTOR), not per pocket site. 1,288 sites collapse to 225 because a homotetramer deposits the same ligand four times. Counting those separately would inflate the R-group frequency prior unevenly -- and that prior is both what Hit@K is judged against and what the logQ correction subtracts, so inflating it moves the number being optimised.

Two historical corrections sit behind the site count. Selecting ligand copies by residue name alone matched backbone residues too: mmCIF records free amino acids as ATOM rather than HETATM, so a `resname TRP` match swept up all 14 tryptophans in the protein -- 2,269 sites, 445 of them 'tryptophan'. And the set was later filtered by SENSOR ORGANISM to drop insect chemoreception; the family labels could not be used for that, since 24 bovine, porcine, canine, rat and human lipocalins were filed under the insect OBP family.

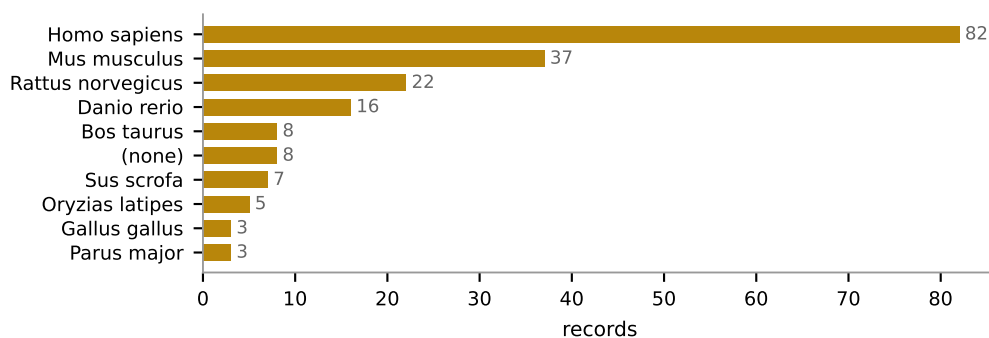
Receptors

what the set actually covers



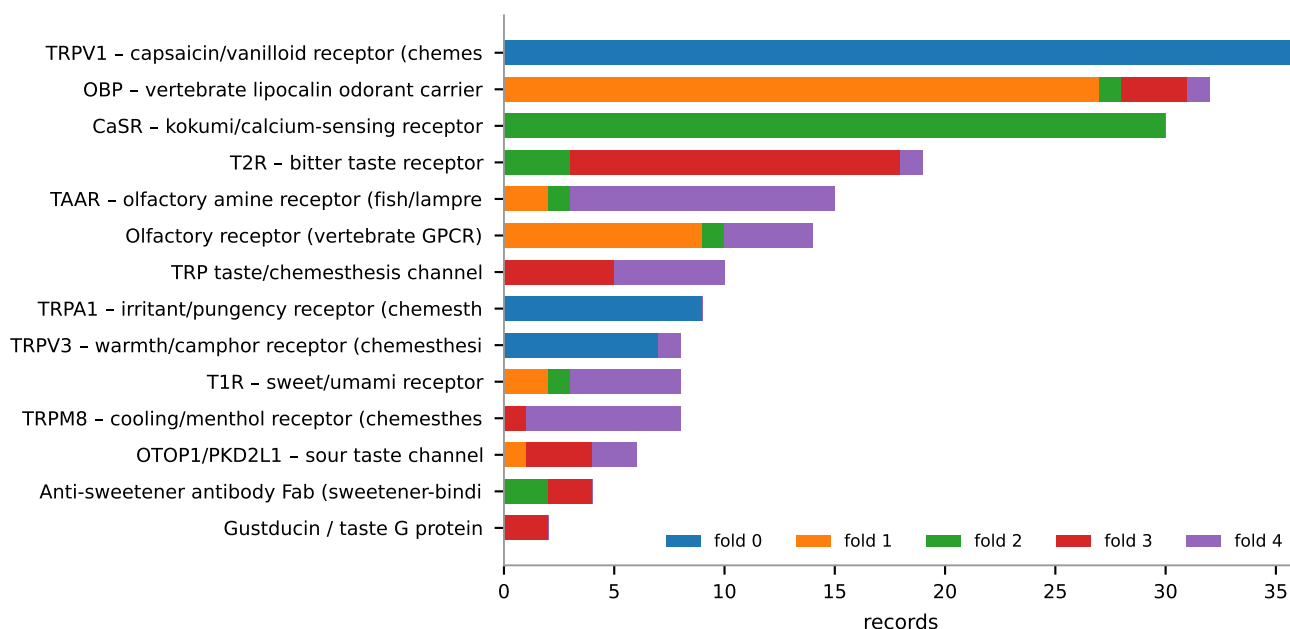
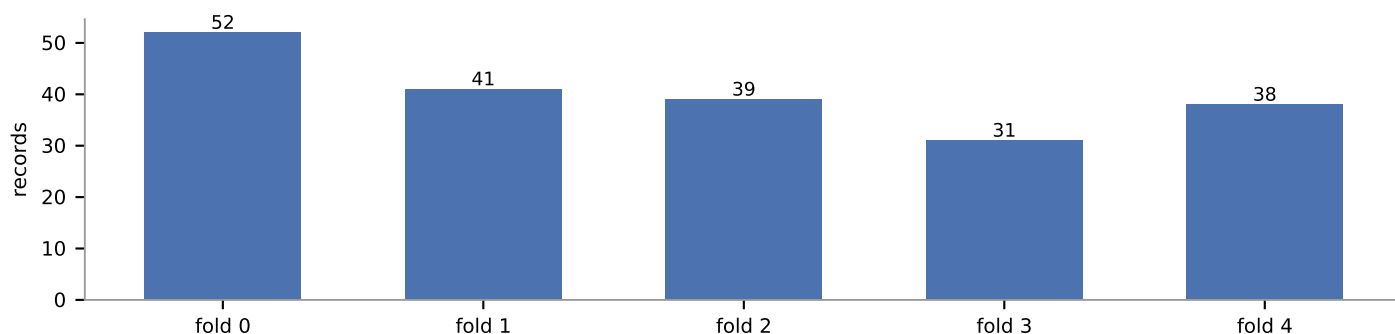
Records and ligands rank differently, and the gap is the point. Highest records-per-ligand: CaSR (3.33), T2R (2.71). Both are amino-acid or bitter sensors where the same few ligands recur across several receptors, and each (ligand, receptor) pair is a separate datapoint.

TRPV1 is the reverse: 30 distinct ligands over 36 records, the widest chemistry in the set. Capsaicinoids, vanilloids and resiniferatoxin analogues -- pungency stimuli whose human descriptors are sparse.



Cross-validation folds

cut on connected components, not at random



A fold holds out whole connected components of the (ligand, receptor) bipartite graph, so a held-out receptor's ligands are absent from training too. Splitting by receptor and patching ligand conflicts afterwards cost 64 of the then-269 records and still left the halves unbalanced.

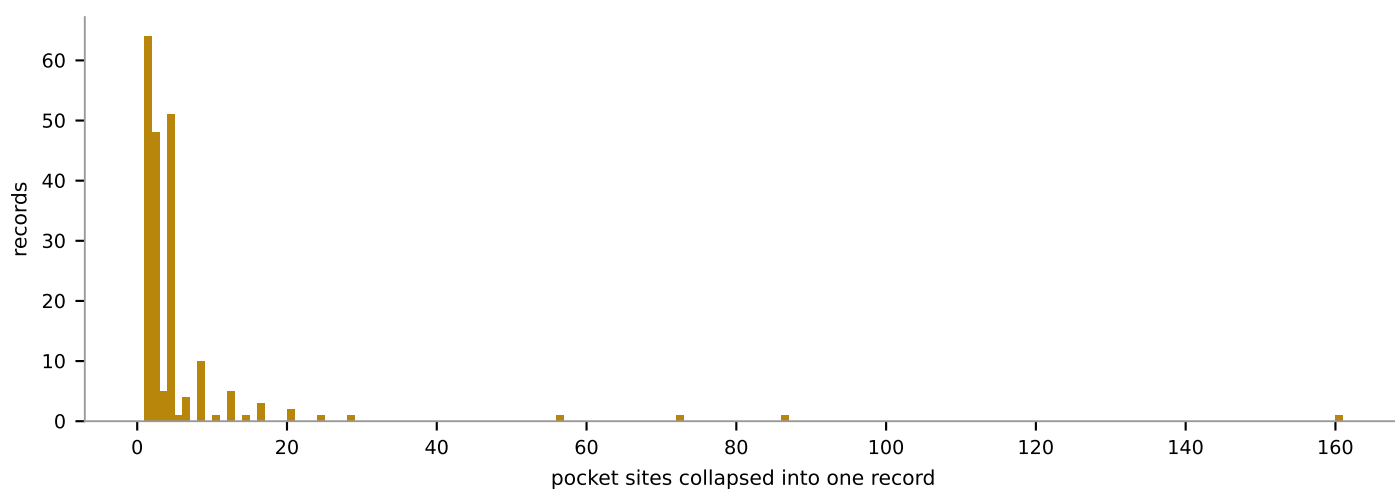
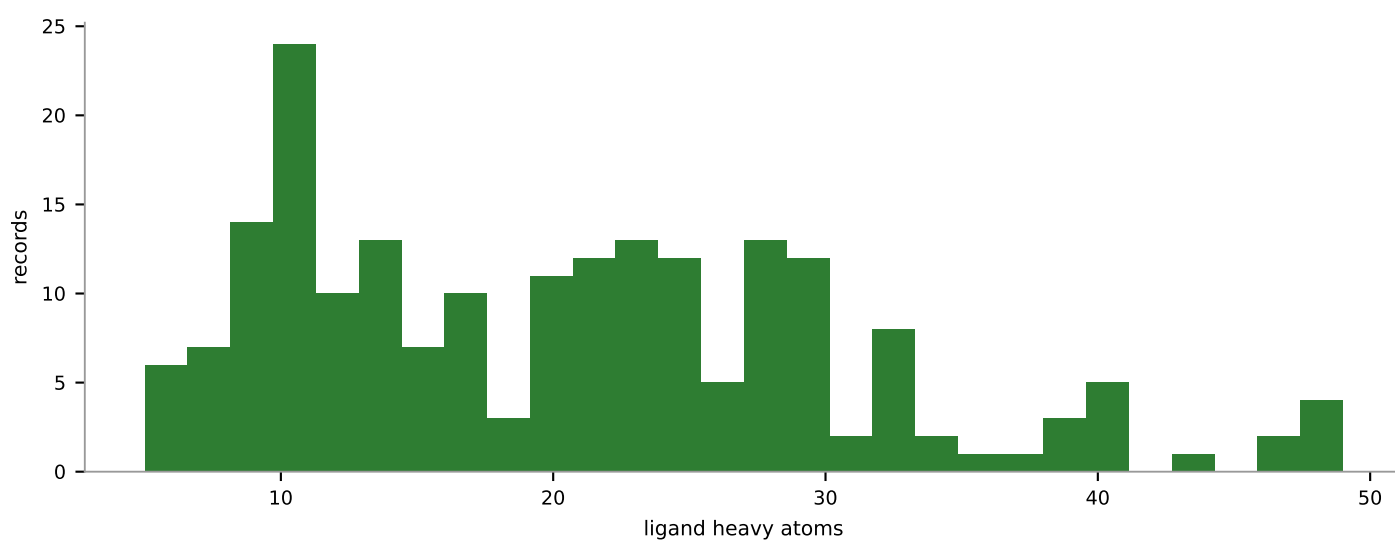
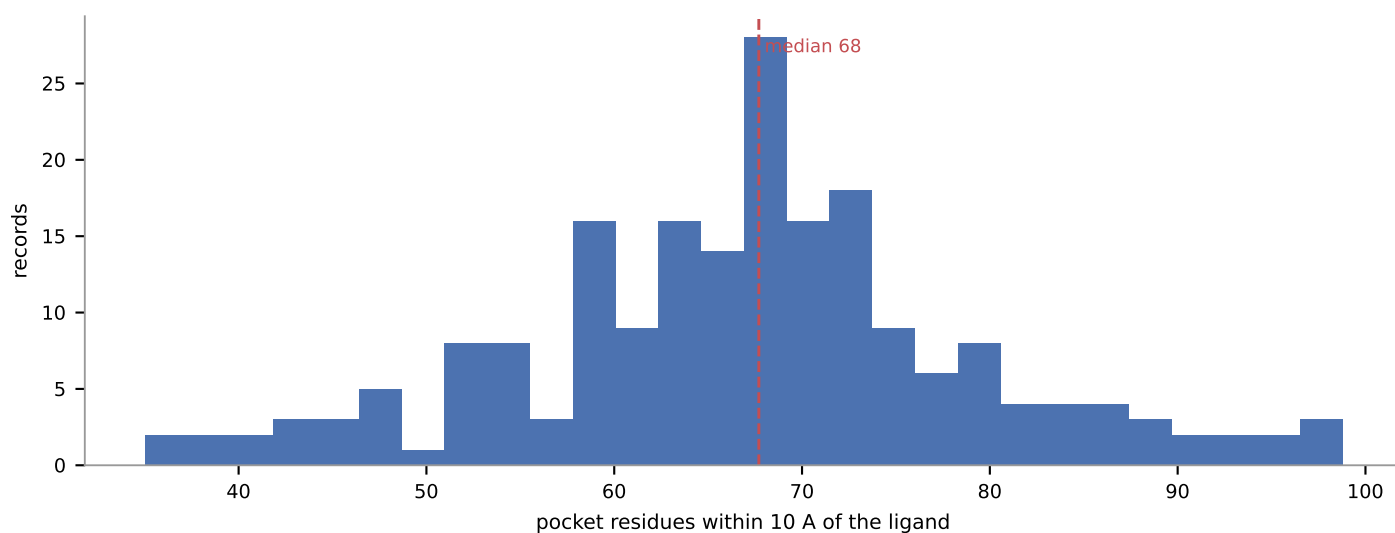
This cannot be random. Pooled ESM-2 pocket embeddings identify the receptor family with 98.5% 1-NN accuracy across different PDB entries, against a 27.4% majority baseline -- a random split puts the same protein on both sides and reports memorisation as generalisation.

4 of 14 families sit in a single fold, because each is one component and cannot be spread at any weight. Those folds test an unseen receptor FAMILY, which is strictly harder than an unseen receptor:

- TRPV1 - capsaicin/vanilloid receptor (chemesthesis)
- CaSR - kokumi/calcium-sensing receptor
- TRPA1 - irritant/pungency receptor (chemesthesis)
- Gustducin / taste G protein

Pockets and ligands

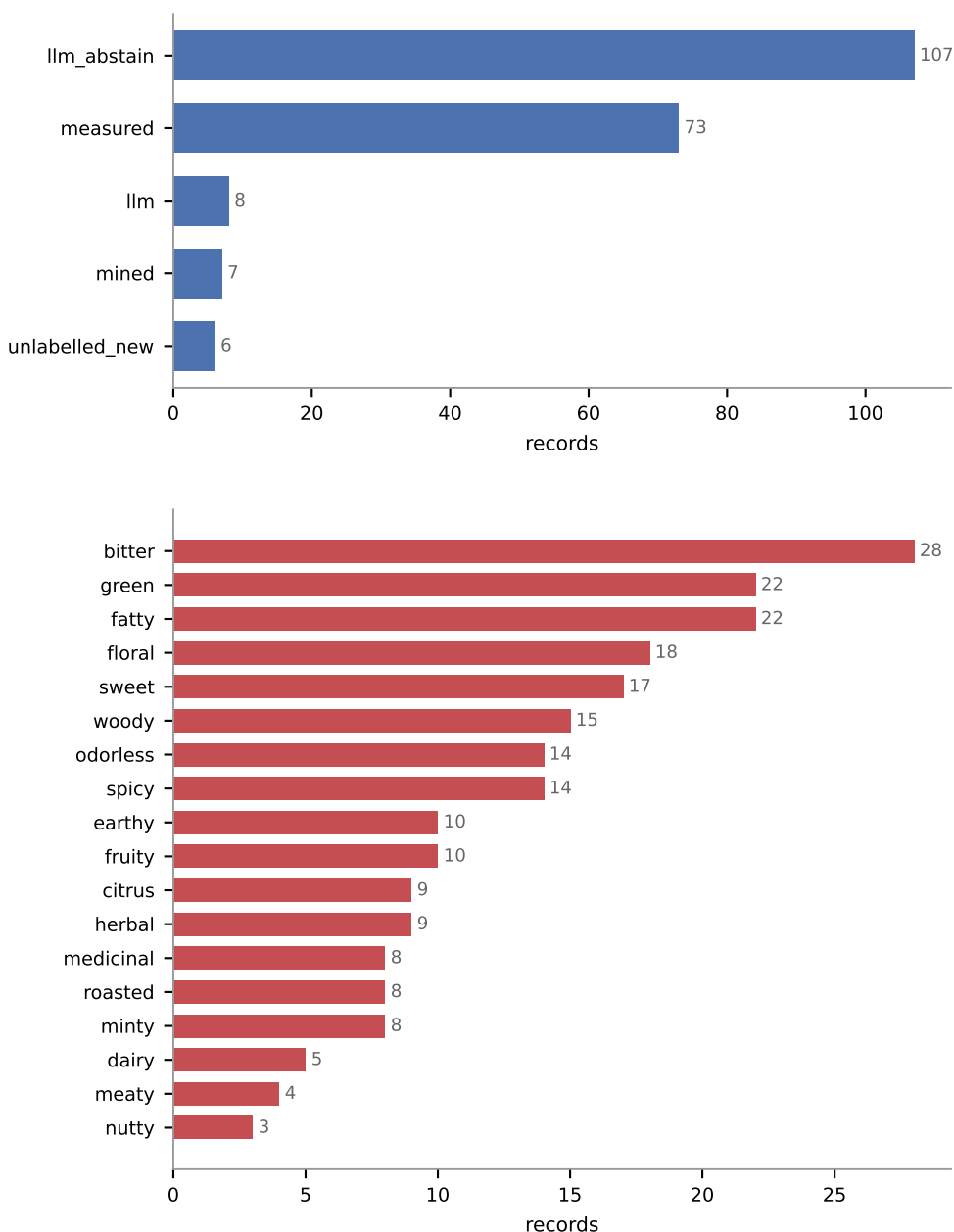
10 Å sites, and the molecules that define them



10 Å matches the pocket10 convention CrossDocked's processed LMDB uses, so pockets from the two sources are directly comparable. Residue selection is 3D even though the embedding is 1D, which is what keeps it usable for GPCRs: in 8F76 the propionate site draws on residues 100-108 and 151-157.

Flavour labels

tables first, then a molecule-only LLM pass



88 of 201 records (44%) carry a real label.

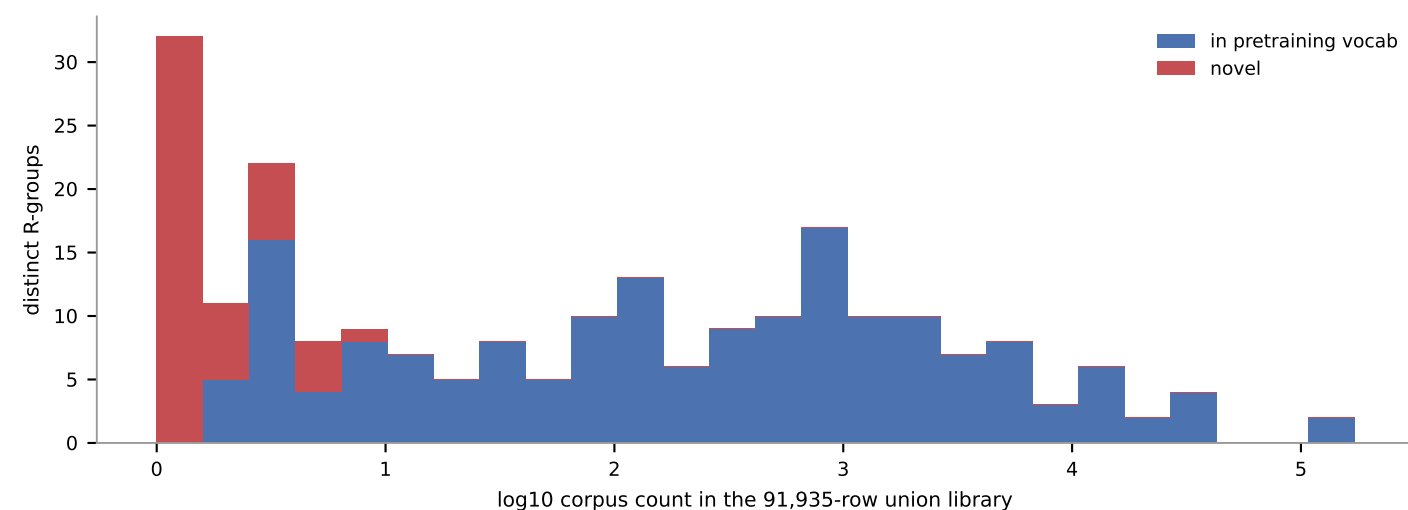
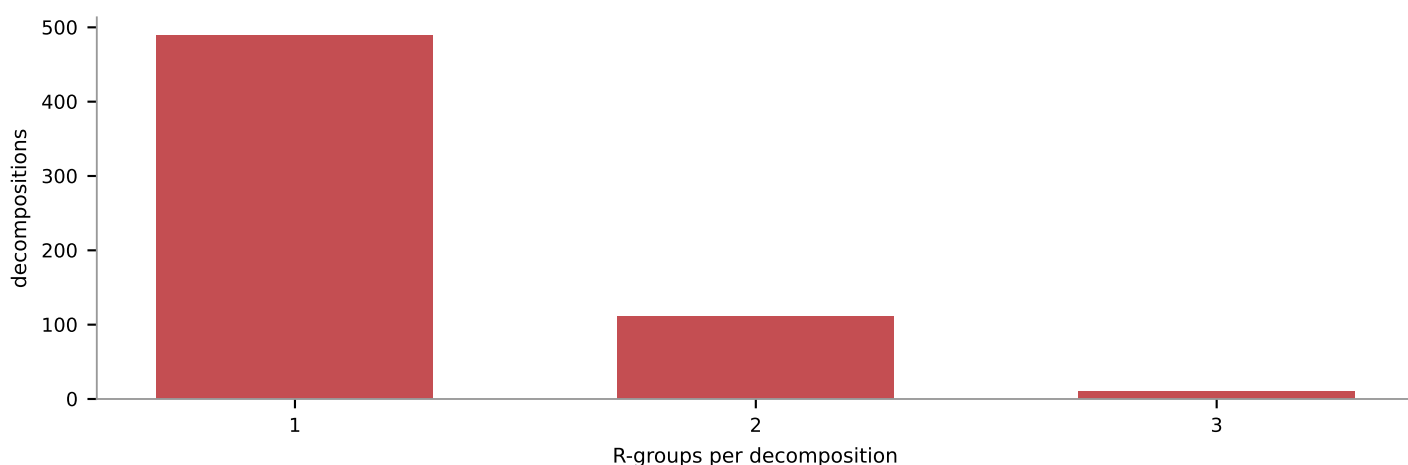
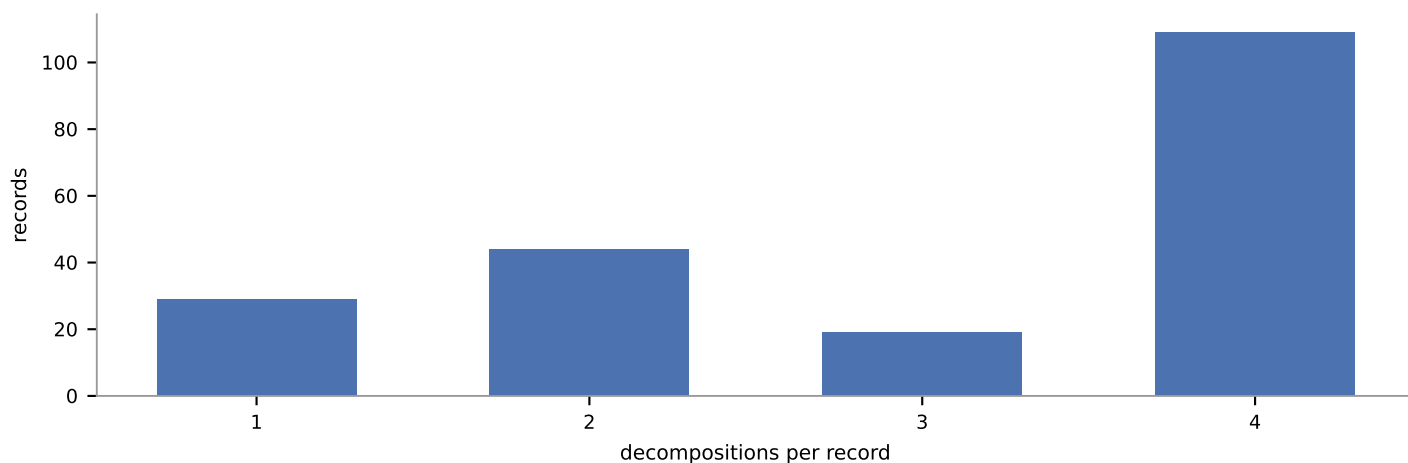
The rest is not a preprocessing failure. Of the molecules the tables could not resolve, most are modulators, lipids or cofactors with no percept at all: synthetic channel agonists, membrane phospholipids from cryo-EM structures, and crystallisation additives that survived filtering. A pocket-bound ligand is not automatically a tastant, and forcing a descriptor onto one would put noise into the half of the convec that demonstrably works.

Insect chemoreception -- previously the largest block here and unlabellable in a human vocabulary by construction -- has been removed from the set.

The LLM pass sees the MOLECULE ONLY -- name and SMILES, never the receptor. Told 'this binds TRPM8' it would answer 'cooling', and the flavour half of the convec would become a re-encoding of the pocket half: conditioning would then show a large lift meaning nothing, because both halves would carry the same variable.

Decompositions

naveja_recap, identical settings to every other corpus



49 of 224 distinct R-groups (22%) are absent from the pretraining vocabulary. They are also RARE: median corpus count 1 against 270 for the rest.

That rarity is why hit@K has to be reported split. Novel R-groups barely move a micro average, so a comfortable headline number is consistent with novel retrieval failing outright. Whether it does is what base_hit@K and novel_hit@K are there to separate -- neither can be read off this chart.

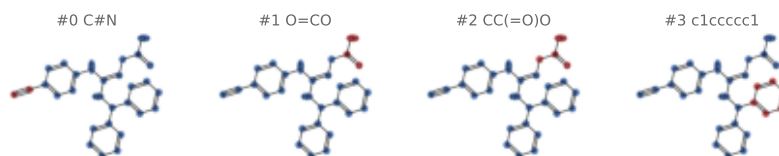
Catalogue

records 1-4 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

GAS · G0YP42;P01867

N-(P-CYANOPHENYL)-N'-DIPHENYLMETHYL-GUANIDINE-ACETIC...

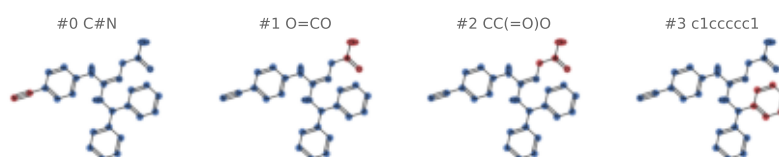
family Anti-sweetener antibody Fab..
organism Mus musculus
PDB 1ETZ
fold 3 sites 2 pocket 54.0 res
flavour (none) [llm_abstain]
SMILES N#Cc1ccc(N/C(=N\CC(=O)O)NC(c2ccccc2)c2cc...



GAS · A2P1G9

N-(P-CYANOPHENYL)-N'-DIPHENYLMETHYL-GUANIDINE-ACETIC...

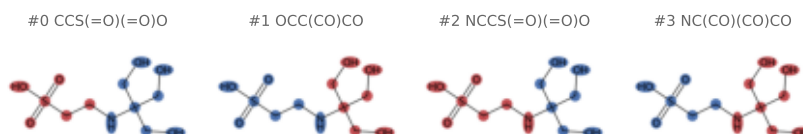
family Anti-sweetener antibody Fab..
organism Mus musculus
PDB 2CGR
fold 3 sites 1 pocket 57.0 res
flavour (none) [llm_abstain]
SMILES N#Cc1ccc(N/C(=N\CC(=O)O)NC(c2ccccc2)c2cc...



NES · A2P1G9;P01867

2-(2-HYDROXY-1,1-DIHYDROXYMETHYL-ETHYLAMINO)-ETHANES...

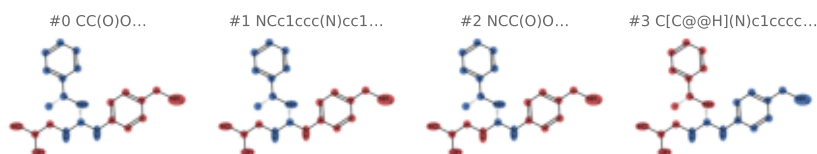
family Anti-sweetener antibody Fab..
organism Mus musculus
PDB 1YNL
fold 2 sites 1 pocket 55.0 res
flavour (none) [llm_abstain]
SMILES O=S(=O)(O)CCNC(CO)(CO)CO



SC5 · A2P1G9;P01867

2-([(R)-{[4-(AMINOMETHYL)PHENYL]AMINO}]{[(1R)-1-PHENY...

family Anti-sweetener antibody Fab..
organism Mus musculus
PDB 1YNK
fold 2 sites 1 pocket 58.0 res
flavour sweet [mined]
SMILES C[C@@H](N[C@@H](NCC(O)O)Nc1ccc(CN)cc1)c1L...



Catalogue

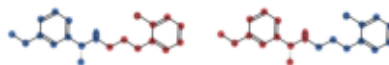
records 5-8 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

9IG · P41180

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 7SIL
fold 2 sites 2 pocket 68.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCC1CCCCC1Cl ● #1 COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1 ●

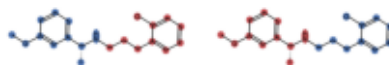


9IG · A0A590UJY2;P41180;P59768;P6287...

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9ASB
fold 2 sites 2 pocket 71.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCC1CCCCC1Cl ● #1 COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1 ●

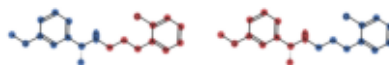


9IG · P41180;P59768;P62873;P63092

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AXF
fold 2 sites 2 pocket 67.5 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCC1CCCCC1Cl ● #1 COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1 ●

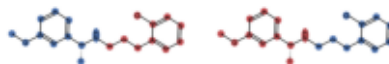


9IG · P41180;P59768;P62873;P63092;P6...

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AVG
fold 2 sites 2 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCC1CCCCC1Cl ● #1 COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1 ●



Catalogue

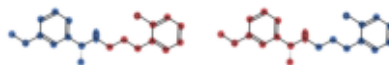
records 9-12 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

9IG · P08754;P41180;P59768;P62879

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AVL
fold 2 sites 2 pocket 67.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCC1CCCCC1Cl ● #1 COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1 ●

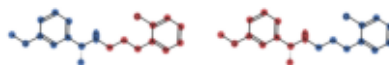


9IG · P41180;P59768;P62873;P63096

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AYF
fold 2 sites 2 pocket 64.5 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCC1CCCCC1Cl ● #1 COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1 ●

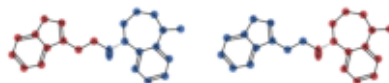


A1ATP · P41180

(5R)-N-[2-(1,2-benzothiazol-3-yl)ethyl]-1-methyl-2,3...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9C1P
fold 2 sites 2 pocket 62.5 res
flavour (none) [llm_abstain]
SMILES CN1CCC[C@@H](NCCc2nsc3ccccc23)c2ccccc21

#0 CCc1nsc2ccccc12 ● #1 CN1CCC[C@@H](N)C1CCC2C(C)CC2 ●

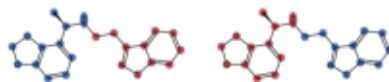


A1ATX · P41180

(1R)-1-(2H-1,3-benzodioxol-4-yl)-N-[2-(1,2-benzothia...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9C2F
fold 2 sites 2 pocket 65.5 res
flavour (none) [llm_abstain]
SMILES C[C@@H](NCCc1nsc2ccccc12)c1cccc2c1OC2

#0 CCc1nsc2ccccc12 ● #1 C[C@@H](N)C1CCCC2C(C)CC2 ●



Catalogue

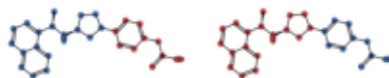
records 13–16 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

H43 · A0A8I3AZV5

2-[4-[(3S)-3-[[[(1R)-1-naphthalen-1-ylethyl]amino]pyr...

family CaSR — kokumi/calcium-sensing receptor
organism Gallus gallus
PDB 7DD7
fold 2 sites 2 pocket 69.0 res
flavour (none) [llm_abstain]
SMILES C[C@@H](N[C@H]1CCN(c2ccc(CC(=O)O)cc2)C1)...

#0 O=C(O)Cc1ccccc1 #1 C[C@@H](N[C@H]1C...

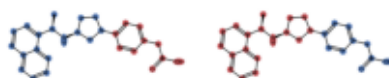


H43 · P41180

2-[4-[(3S)-3-[[[(1R)-1-naphthalen-1-ylethyl]amino]pyr...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 7M3G
fold 2 sites 2 pocket 67.0 res
flavour (none) [llm_abstain]
SMILES C[C@@H](N[C@H]1CCN(c2ccc(CC(=O)O)cc2)C1)...

#0 O=C(O)Cc1ccccc1 #1 C[C@@H](N[C@H]1C...



SPM · P41180

SPERMINE

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZF
fold 2 sites 1 pocket 47.0 res
flavour (none) [llm_abstain]
SMILES NCCCCCNCNCCN

#0 CCCN CCCN #1 CCCN NCCCN #2 NCCCN CCCN #3 CCCN



SPM · P08754;P41180;P59768;P62873

SPERMINE

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZH
fold 2 sites 4 pocket 37.5 res
flavour (none) [llm_abstain]
SMILES NCCCCCNCNCCN

#0 CCCN CCCN #1 CCCN NCCCN #2 NCCCN CCCN #3 CCCN



Catalogue

records 17–20 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

SPM · P41180;P50148;P59768;P62873;P6...

SPERMINE

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZG
fold 2 sites 3 pocket 35.0 res
flavour (none) [llm_abstain]
SMILES NCCCNCCCNCNCCN

#0 CCCN CCCN



#1 CCCN NCCCN



#2 NCCCN CCCN



#3 CCCN

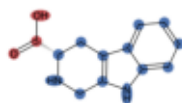


TCR · P41180;P59768;P62873;P63096

CYCLOMETHYLTRYPTOPHAN

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AYF
fold 2 sites 2 pocket 63.5 res
flavour (none) [llm_abstain]
SMILES O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1

#0 O=CO

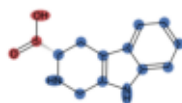


TCR · A0A590UJY2;P41180;P59768;P6287...

CYCLOMETHYLTRYPTOPHAN

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9ASB
fold 2 sites 2 pocket 70.0 res
flavour (none) [llm_abstain]
SMILES O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1

#0 O=CO

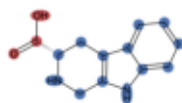


TCR · P41180;P59768;P62873;P63092;P6...

CYCLOMETHYLTRYPTOPHAN

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AVG
fold 2 sites 2 pocket 71.0 res
flavour (none) [llm_abstain]
SMILES O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1

#0 O=CO



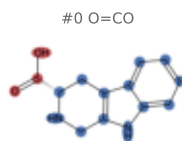
Catalogue

records 21–24 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

TCR · P41180

CYCLOMETHYLTRYPTOPHAN

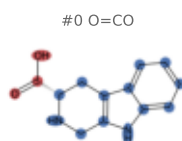
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 5FBH 5FBK 7E6T 7SIL 7SIM
fold 2 sites 10 pocket 67.9 res
flavour (none) [llm_abstain]
SMILES O=C(O)[C@H]1Cc2c([nH]c3ccccc23)CN1



TCR · P08754;P41180;P59768;P62879

CYCLOMETHYLTRYPTOPHAN

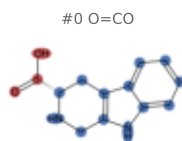
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AVL
fold 2 sites 2 pocket 66.5 res
flavour (none) [llm_abstain]
SMILES O=C(O)[C@H]1Cc2c([nH]c3ccccc23)CN1



TCR · P41180;P59768;P62873;P63092

CYCLOMETHYLTRYPTOPHAN

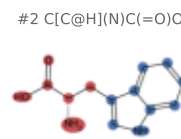
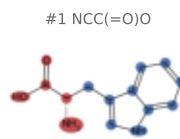
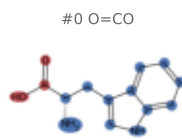
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AXF
fold 2 sites 2 pocket 65.0 res
flavour (none) [llm_abstain]
SMILES O=C(O)[C@H]1Cc2c([nH]c3ccccc23)CN1



TRP · P41180;P50148;P59768;P62873;P6...

TRYPTOPHAN

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZG
fold 2 sites 2 pocket 67.5 res
flavour bitter [measured]
SMILES N[C@H](Cc1c[nH]c2ccccc12)C(=O)O



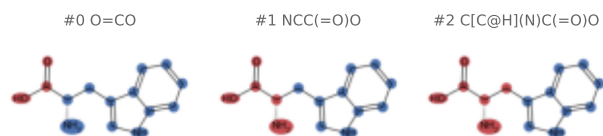
Catalogue

records 25–28 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

TRP · P08754;P41180;P59768;P62873

TRYPTOPHAN

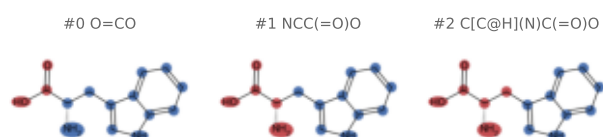
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZH 8SZI
fold 2 sites 5 pocket 64.4 res
flavour bitter [measured]
SMILES N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O



TRP · P41180

TRYPTOPHAN

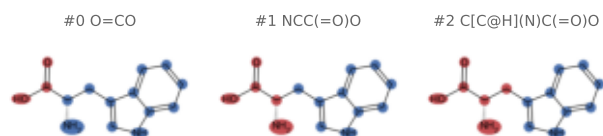
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 5K5S 7DTU 7DTV 7M3E 7M3F 7M3G 8SZF 8WPG..
fold 2 sites 20 pocket 67.5 res
flavour bitter [measured]
SMILES N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O



TRP · P04899;P41180;P59768;P62873;P6...

TRYPTOPHAN

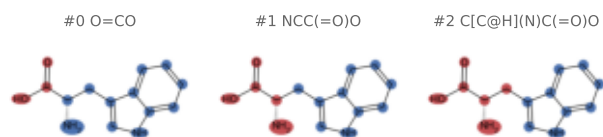
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8WPU
fold 2 sites 2 pocket 71.5 res
flavour bitter [measured]
SMILES N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O



TRP · A0A8I3AZV5

TRYPTOPHAN

family CaSR — kokumi/calcium-sensing receptor
organism Gallus gallus
PDB 7DD5 7DD6 7DD7
fold 2 sites 6 pocket 67.8 res
flavour bitter [measured]
SMILES N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O



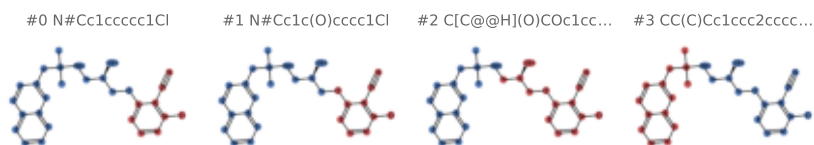
Catalogue

records 29–32 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

YP1 · P41180

2-chloro-6-[(2R)-2-hydroxy-3-{[2-methyl-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromethyl)ethyl]-3-(trifluoromethyl)ethyl]amino}propanoic acid]

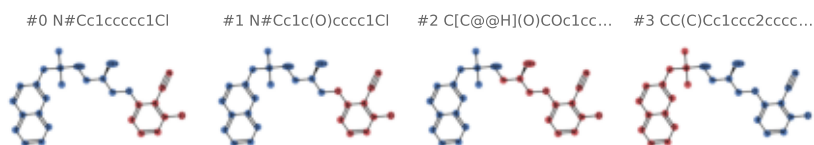
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 7M3E 7M3J 7SIN
fold 2 sites 6 pocket 68.5 res
flavour (none) [llm_abstain]
SMILES CC(C)(Cc1ccc2ccccc2c1)NC[C@@H](O)C(=O)Cc1ccc(Cl)cc1



YP1 · A0A8I3AZV5

2-chloro-6-[(2R)-2-hydroxy-3-{[2-methyl-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromethyl)ethyl]-3-(trifluoromethyl)ethyl]amino}propanoic acid]

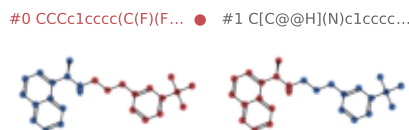
family CaSR — kokumi/calcium-sensing receptor
organism Gallus gallus
PDB 7DD5
fold 2 sites 2 pocket 62.0 res
flavour (none) [llm_abstain]
SMILES CC(C)(Cc1ccc2ccccc2c1)NC[C@@H](O)C(=O)Cc1ccc(Cl)cc1



YP4 · P08754;P41180;P59768;P62873

N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromethyl)ethyl]-3-(trifluoromethyl)ethyl]amino

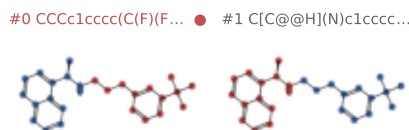
family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZH
fold 2 sites 2 pocket 61.5 res
flavour (none) [llm_abstain]
SMILES C[C@@H](NCCCC1CCCC(C(F)(F)F)C1)C1CCCC2CC(C(F)(F)F)CC2



YP4 · P04899;P41180;P59768;P62873;P62873

N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromethyl)ethyl]-3-(trifluoromethyl)ethyl]amino

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8WPU
fold 2 sites 2 pocket 69.0 res
flavour (none) [llm_abstain]
SMILES C[C@@H](NCCCC1CCCC(C(F)(F)F)C1)C1CCCC2CC(C(F)(F)F)CC2



Catalogue

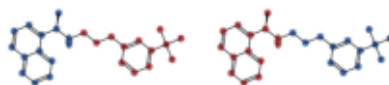
records 33–36 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

YP4 · P41180;P50148;P59768;P62873;P6...

N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromet...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 8SZG
fold 2 sites 2 pocket 64.0 res
flavour (none) [llm_abstain]
SMILES C[C@@H](NCCCC1CCCC(C(F)(F)F)C1)C1CCCC2CC...

#0 CCCC1CCCC(C(F)(F)F)C1 ● #1 C[C@@H](N)C1CCCC...

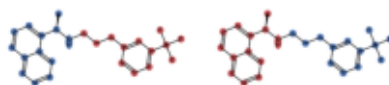


YP4 · P41180

N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromet...

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 7M3F 8SZF 8WPG
fold 2 sites 6 pocket 64.5 res
flavour (none) [llm_abstain]
SMILES C[C@@H](NCCCC1CCCC(C(F)(F)F)C1)C1CCCC2CC...

#0 CCCC1CCCC(C(F)(F)F)C1 ● #1 C[C@@H](N)C1CCCC...



A1AEI · A8MTJ3;P59768;P62873;Q59310;Q9...

4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

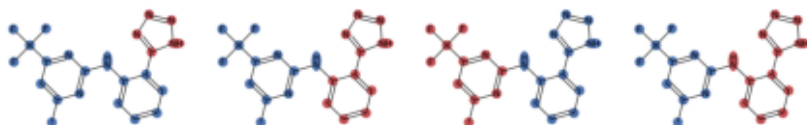
family Gustducin / taste G protein
organism Homo sapiens
PDB 8YKY
fold 3 sites 1 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n...

#0 c1nnn[nH]1

#1 c1ccc(-c2nnn[nH]2)cc1

#2 Cc1cc(C(F)(F)F)n... ●

#3 Nc1ccccc1-c1nnn[nH]1



U9I · P41145;P59768;P62873

methyl...

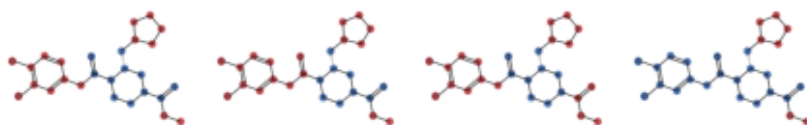
family Gustducin / taste G protein
organism Homo sapiens
PDB 8DZR
fold 3 sites 1 pocket 88.0 res
flavour (none) [llm_abstain]
SMILES COC(=O)N1CCN(C(=O)Cc2ccc(Cl)c(Cl)c2)[C@H]...

#0 CO C1CCNC1...

#1 CO...

#2 COC=O C1CCNC1...

#3 CO C1CCNC1



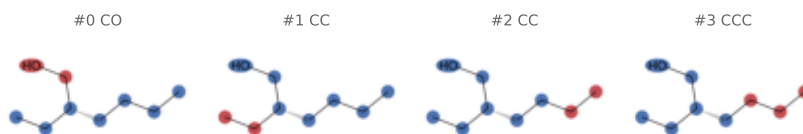
Catalogue

records 37–40 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

2EH · P11590

(2S)-2-ethylhexan-1-ol

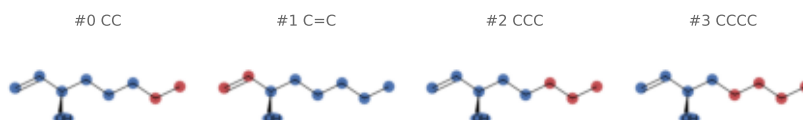
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 3KFH
fold 3 sites 1 pocket 61.0 res
flavour citrus; fatty; floral; green; swee.. [measured]
SMILES CCCC[C@H](CC)CO



3OL · P07435

(3R)-oct-1-en-3-ol

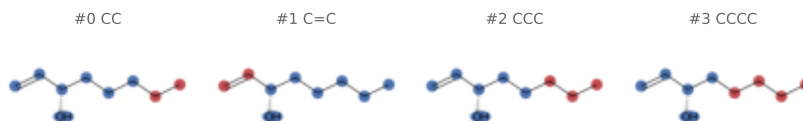
family OBP – vertebrate lipocalin odorant..
organism *Bos taurus*
PDB 1G85
fold 1 sites 2 pocket 65.5 res
flavour earthy; fatty; green; meaty; odorL.. [measured]
SMILES C=C[C@H](O)CCCC



3OM · P07435

(3S)-1-octen-3-ol

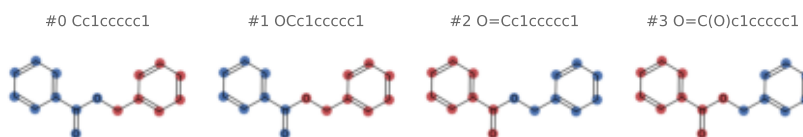
family OBP – vertebrate lipocalin odorant..
organism *Bos taurus*
PDB 1GT1 1GT3
fold 1 sites 3 pocket 65.0 res
flavour earthy; fatty; green; meaty; odorL.. [measured]
SMILES C=C[C@@H](O)CCCC



BZM · P81245

BENZOIC ACID PHENYLMETHYLESTER

family OBP – vertebrate lipocalin odorant..
organism *Sus scrofa*
PDB 1DZM
fold 1 sites 2 pocket 73.0 res
flavour dairy; fatty; fruity; herbal; nutt.. [measured]
SMILES O=C(Oc1ccccc1)c1ccccc1



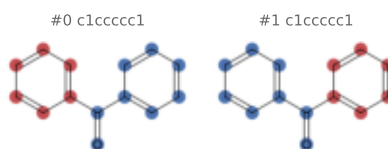
Catalogue

records 41–44 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

BZQ · P81245

DIPHENYLMETHANONE

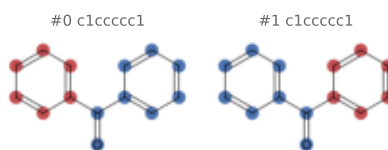
family OBP – vertebrate lipocalin odorant..
organism *Sus scrofa*
PDB 1DZP
fold 1 sites 2 pocket 68.5 res
flavour floral;fruity;medicinal;woody [measured]
SMILES O=C(c1ccccc1)c1ccccc1



BZQ · P07435

DIPHENYLMETHANONE

family OBP – vertebrate lipocalin odorant..
organism *Bos taurus*
PDB 1GT5
fold 1 sites 2 pocket 70.0 res
flavour floral;fruity;medicinal;woody [measured]
SMILES O=C(c1ccccc1)c1ccccc1



DE1 · P11589

DECAN-1-OL

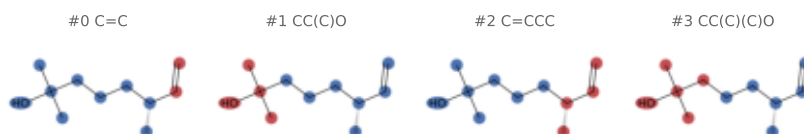
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1ZNL
fold 1 sites 1 pocket 77.0 res
flavour citrus;fatty;floral;odorless;s.. [measured]
SMILES CCCCCCCCCO



DHM · P07435

2,6-DIMETHYL-7-OCTEN-2-OL

family OBP – vertebrate lipocalin odorant..
organism *Bos taurus*
PDB 1GT3
fold 1 sites 2 pocket 66.0 res
flavour citrus;floral;green;odorless [measured]
SMILES C=C[C@H](C)CCCC(C)(C)O



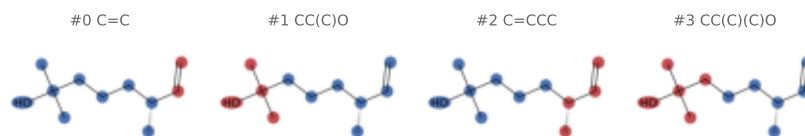
Catalogue

records 45–48 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

DHM · P81245

2,6-DIMETHYL-7-OCTEN-2-OL

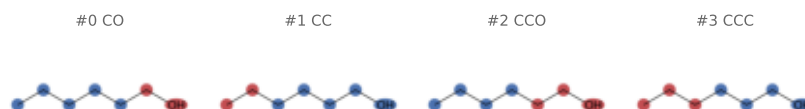
family OBP – vertebrate lipocalin odorant..
organism *Sus scrofa*
PDB 1E00
fold 1 sites 2 pocket 72.5 res
flavour citrus;floral;green;odorless [measured]
SMILES C=C[C@H](C)CCCC(C)(C)O



HE2 · P11589

HEXAN-1-OL

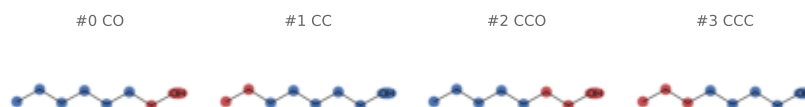
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1ZNE
fold 1 sites 1 pocket 60.0 res
flavour alcoholic;fatty;floral;fruity;... [measured]
SMILES CCCCCCO



HE4 · P11589

HEPTAN-1-OL

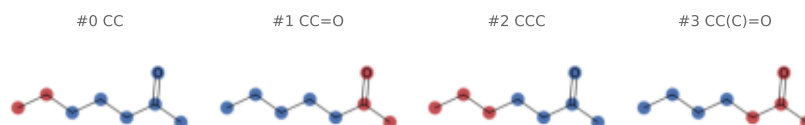
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1ZNG
fold 1 sites 1 pocket 72.0 res
flavour earthy;floral;fruity;green;her... [measured]
SMILES CCCCCCC



HTX · P11590

heptan-2-one

family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 3KFG
fold 3 sites 1 pocket 59.0 res
flavour fatty;fruity;herbal;nutty;spic... [measured]
SMILES CCCCCC(=O)



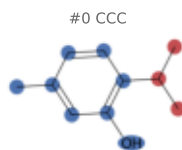
Catalogue

records 49-52 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

IPB · P81245

5-METHYL-2-(1-METHYLETHYL)PHENOL

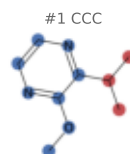
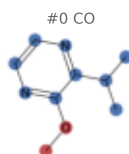
family OBP – vertebrate lipocalin odorant..
organism *Sus scrofa*
PDB 1E06
fold 1 sites 2 pocket 66.5 res
flavour herbal;medicinal;woody [measured]
SMILES Cc1ccc(C(C)C)c(O)c1



IPZ · P11589

2-ISOPROPYL-3-METHOXPYRAZINE

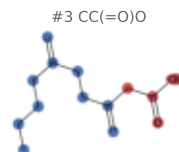
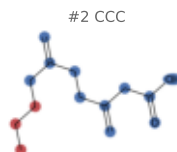
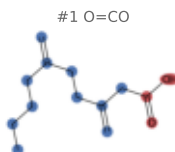
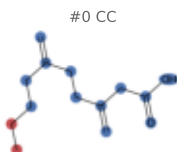
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1QY2
fold 1 sites 1 pocket 63.0 res
flavour earthy [measured]
SMILES COc1nccn1C(C)C



LIK · P07435

3,6-BIS(METHYLENE)DECANOIC ACID

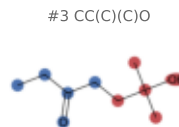
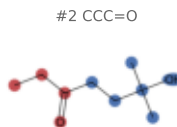
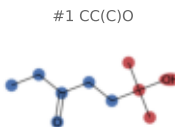
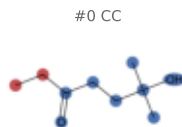
family OBP – vertebrate lipocalin odorant..
organism *Bos taurus*
PDB 2HLV
fold 1 sites 1 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES C=C(CCCC)CCC(=C)CC(=O)O



LTL · P02762

6-HYDROXY-6-METHYL-HEPTAN-3-ONE

family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1I05
fold 1 sites 1 pocket 67.0 res
flavour (none) [llm_abstain]
SMILES CCC(=O)CCC(C)(C)O



Catalogue

records 53–56 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

LWE · H2B3G5

(+)-menthone

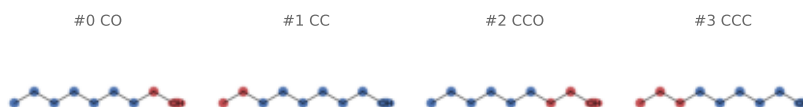
family OBP – vertebrate lipocalin odorant..
organism *Canis lupus familiaris*
PDB 8AEJ
fold 2 sites 1 pocket 65.0 res
flavour minty [measured]
SMILES CC(C)[C@H]1CC[C@H](C)CC1=O



OC9 · P11589

OCTAN-1-OL

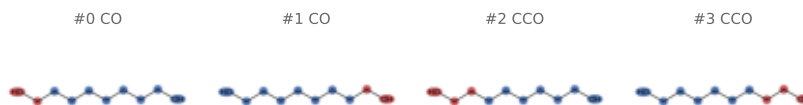
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1ZNH
fold 1 sites 1 pocket 75.0 res
flavour citrus;earthy;fatty;floral;gre.. [measured]
SMILES CCCCCCCCO



ODI · P11588

OCTANE-1,8-DIOL

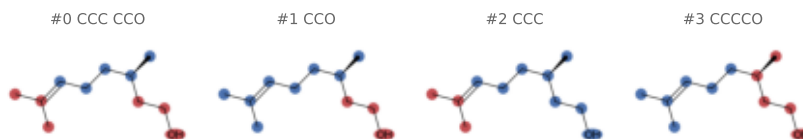
family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 2DM5
fold 1 sites 1 pocket 77.0 res
flavour (none) [llm_abstain]
SMILES OCCCCCCCCO



ODM · O18874

(3R)-3,7-dimethyloct-6-en-1-ol

family OBP – vertebrate lipocalin odorant..
organism *Canis lupus familiaris*
PDB 8AEI
fold 4 sites 2 pocket 76.5 res
flavour bitter;fatty;floral;green [measured]
SMILES CC(C)=CCC[C@H](C)CCO



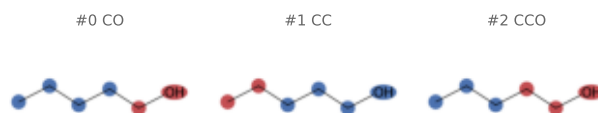
Catalogue

records 57–60 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PE9 · P11589

PENTAN-1-OL

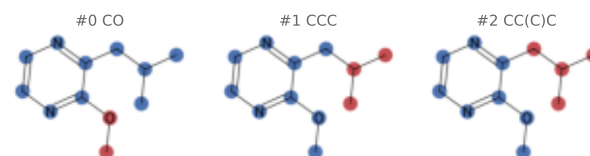
family OBP – vertebrate lipocalin odorant..
organism Mus musculus
PDB 1ZND
fold 1 sites 2 pocket 59.5 res
flavour alcoholic;fatty;sweet;woody [measured]
SMILES CCCCCO



PRZ · P11588

2-ISOBUTYL-3-METHOXPYRAZINE

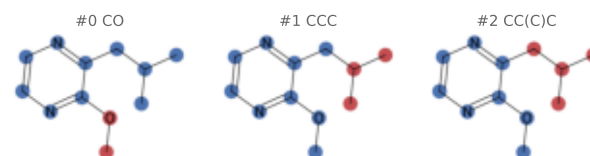
family OBP – vertebrate lipocalin odorant..
organism Mus musculus
PDB 1QY1 1YP6
fold 1 sites 2 pocket 70.0 res
flavour earthy;green;spicy [measured]
SMILES COc1nccnc1CC(C)C



PRZ · P11589

2-ISOBUTYL-3-METHOXPYRAZINE

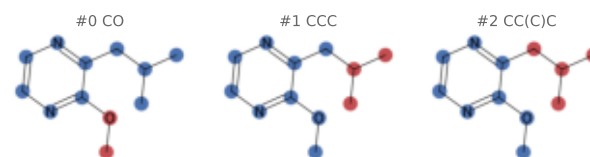
family OBP – vertebrate lipocalin odorant..
organism Mus musculus
PDB 2NND
fold 1 sites 1 pocket 69.0 res
flavour earthy;green;spicy [measured]
SMILES COc1nccnc1CC(C)C



PRZ · P07435

2-ISOBUTYL-3-METHOXPYRAZINE

family OBP – vertebrate lipocalin odorant..
organism Bos taurus
PDB 1GT1
fold 1 sites 1 pocket 71.0 res
flavour earthy;green;spicy [measured]
SMILES COc1nccnc1CC(C)C



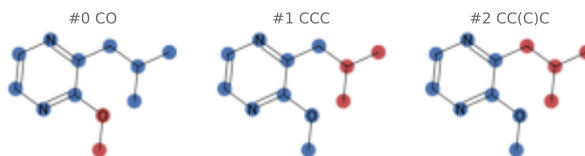
Catalogue

records 61–64 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PRZ · P81245

2-ISOBUTYL-3-METHOXPYRAZINE

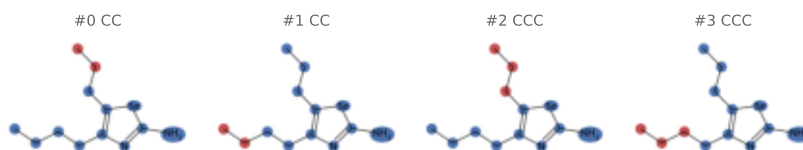
family OBP – vertebrate lipocalin odorant..
organism *Sus scrofa*
PDB 1DZK 1HQP
fold 1 sites 3 pocket 68.0 res
flavour earthy;green;spicy [measured]
SMILES COc1nccnc1CC(C)C



SES · P07435

4-butyl-5-propyl-1,3-selenazol-2-amine

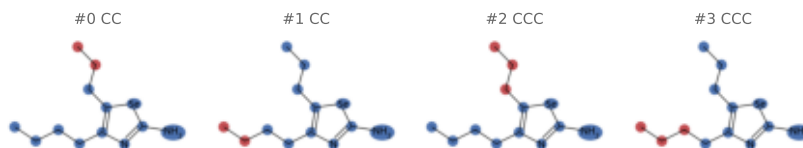
family OBP – vertebrate lipocalin odorant..
organism *Bos taurus*
PDB 1PBO
fold 1 sites 2 pocket 69.5 res
flavour (none) [llm_abstain]
SMILES CCCC1nc(N)[se]c1CCC



SES · P81245

4-butyl-5-propyl-1,3-selenazol-2-amine

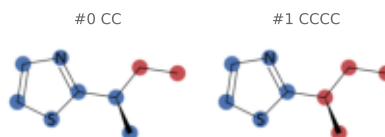
family OBP – vertebrate lipocalin odorant..
organism *Sus scrofa*
PDB 1DZJ
fold 1 sites 2 pocket 71.5 res
flavour (none) [llm_abstain]
SMILES CCCC1nc(N)[se]c1CCC



TZL · P11589

2-(SEC-BUTYL)THIAZOLE

family OBP – vertebrate lipocalin odorant..
organism *Mus musculus*
PDB 1JV4
fold 1 sites 1 pocket 67.0 res
flavour green;herbal;odorless;spicy [measured]
SMILES CC[C@H](C)c1nccs1



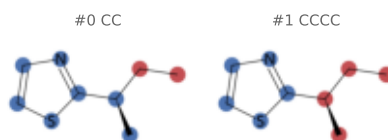
Catalogue

records 65–68 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

TZL · P02762

2-(SEC-BUTYL)THIAZOLE

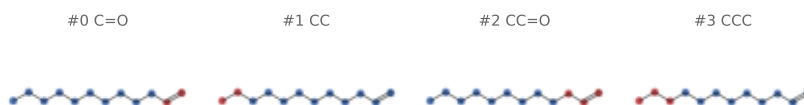
family OBP – vertebrate lipocalin odorant..
organism Mus musculus
PDB 1I06 1MUP
fold 1 sites 2 pocket 64.5 res
flavour green;herbal;odorless;spicy [measured]
SMILES CC[C@H](C)c1nccs1



UNA · P07435

UNDECANAL

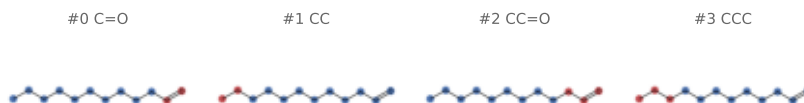
family OBP – vertebrate lipocalin odorant..
organism Bos taurus
PDB 1GT4
fold 1 sites 2 pocket 65.0 res
flavour citrus;fatty;floral;green;spic.. [measured]
SMILES CCCCCCCCC=O



UNA · P81245

UNDECANAL

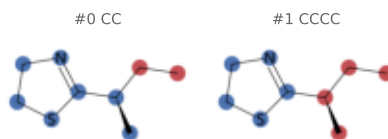
family OBP – vertebrate lipocalin odorant..
organism Sus scrofa
PDB 1E02
fold 1 sites 2 pocket 69.0 res
flavour citrus;fatty;floral;green;spic.. [measured]
SMILES CCCCCCCCC=O



XBT · P11590

2-[(1R)-1-methylpropyl]-4,5-dihydro-1,3-thiazole

family OBP – vertebrate lipocalin odorant..
organism Mus musculus
PDB 3KFF
fold 3 sites 1 pocket 63.0 res
flavour sulfurous [llm]
SMILES CC[C@H](C)C1=NCCS1



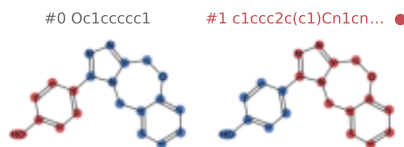
Catalogue

records 69-72 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1BKT · Q7ZWK8

4-[(4R)-5H,11H-[1,2,4]triazolo[3,4-c][1,4]benzoxazep...

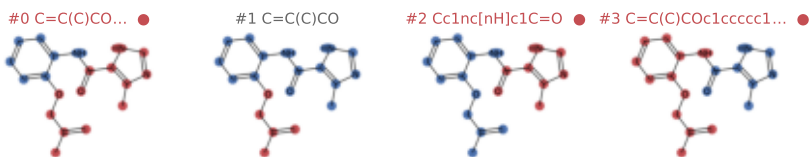
family OTOP1/PKD2L1 — sour taste channel
organism Danio rerio
PDB 9MFM
fold 3 sites 6 pocket 58.7 res
flavour (none) [llm_abstain]
SMILES Oc1ccc(-c2nnc3n2Cc2ccccc2OC3)cc1



A1BKU · Q7ZWK8

4-methyl-N-{2-[(2-methylprop-2-en-1-yl)oxy]phenyl}-1...

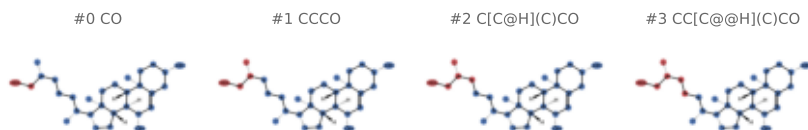
family OTOP1/PKD2L1 — sour taste channel
organism Danio rerio
PDB 9MFF
fold 3 sites 4 pocket 55.0 res
flavour (none) [llm_abstain]
SMILES C=C(C)C0c1ccccc1NC(=O)c1[nH]cnc1C



A1EQ3 · P98161;Q13563

(3~{S},7~{R},8~{S},9~{S},10~{R},13~{R},14~{S},17~{R})...

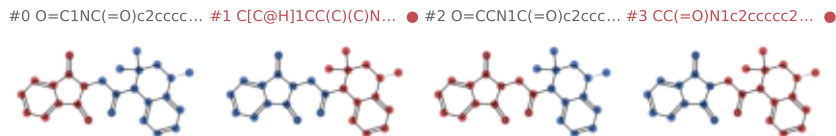
family OTOP1/PKD2L1 — sour taste channel
organism Homo sapiens
PDB 8ZL8
fold 4 sites 1 pocket 65.0 res
flavour (none) [unlabelled_new]
SMILES C[C@@H](C(=O)CCC[C@@H](C)[C@H]1CC[C@H]2[C@H]



AQV · Q13563

2-{2-oxo-2-[(4S)-2,2,4-trimethyl-3,4-dihydroquinolin...

family OTOP1/PKD2L1 — sour taste channel
organism Homo sapiens
PDB 8HK7
fold 1 sites 4 pocket 62.8 res
flavour (none) [unlabelled_new]
SMILES C[C@H]1CC(C)(C)N(C(=O)CN2C(=O)c3ccccc3C2.



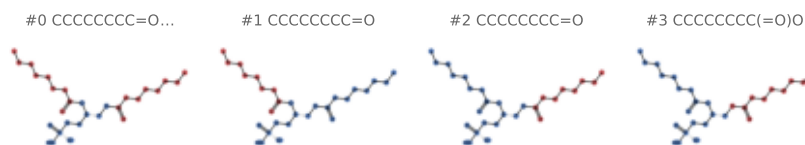
Catalogue

records 73–76 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PA8 · P98161;Q13563

1,2-DIOCTANOYL-SN-GLYCERO-3-PHOSPHATE

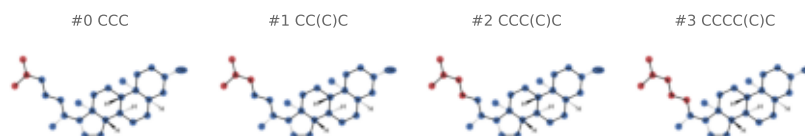
family OTOP1/PKD2L1 – sour taste channel
organism Homo sapiens
PDB 8Z6B 8ZKR
fold 4 sites 2 pocket 82.0 res
flavour (none) [unlabelled_new]
SMILES CCCCCCC(=O)OC[C@H](CO[P@@](=O)([O-])O)O.



QNJ · Q7ZWK8

(3beta,5beta,14beta,17alpha)-cholestan-3-ol

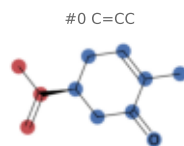
family OTOP1/PKD2L1 – sour taste channel
organism Danio rerio
PDB 9MFL
fold 3 sites 4 pocket 40.5 res
flavour (none) [llm_abstain]
SMILES CC(C)CCC[C@@H](C)[C@H]1CC[C@H]2[C@H]3CC..



OWU · P59768;P62873;P63092

(5S)-2-methyl-5-(prop-1-en-2-yl)cyclohex-2-en-1-one

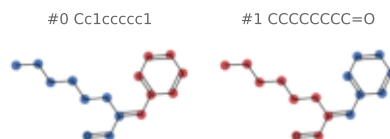
family Olfactory receptor (vertebrate GPCR)
organism
PDB 8UY0
fold 4 sites 1 pocket 63.0 res
flavour roasted;spicy [measured]
SMILES C=C(C)[C@H]1CC=C(C)C(=O)C1



A1EJG · P38405;P59768;P62873

(2~{E})-2-(phenylmethylidene)octanal

family Olfactory receptor (vertebrate GPCR)
organism
PDB 9LDW
fold 1 sites 1 pocket 83.0 res
flavour fatty;floral;green;herbal [measured]
SMILES CCCCC/C(C=O)=C/c1ccccc1



Catalogue

records 77-80 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1EJH · P38405;P59768;P62873

(~{E})-dec-2-enal

family Olfactory receptor (vertebrate GPCR)
organism
PDB 9LE1 9LE2
fold 1 sites 2 pocket 68.0 res
flavour citrus;earthy;fatty;floral;gre. [measured]
SMILES CCCCCC/C=C/C=O

#0 CC=O

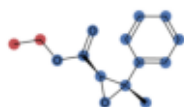


A1EJQ · P59768;P62873

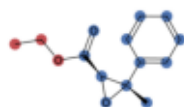
ethyl...

family Olfactory receptor (vertebrate GPCR)
organism
PDB 9LDZ
fold 1 sites 1 pocket 68.0 res
flavour fatty;floral;fruity;sweet [measured]
SMILES CCOC(=O)[C@@H]10[C@@]1(C)C1CCCC1

#0 CC



#1 CCO

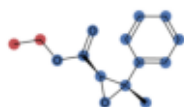


A1EJQ · P38405;P59768;P62873

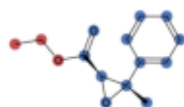
ethyl...

family Olfactory receptor (vertebrate GPCR)
organism
PDB 9LDV
fold 1 sites 1 pocket 70.0 res
flavour fatty;floral;fruity;sweet [measured]
SMILES CCOC(=O)[C@@H]10[C@@]1(C)C1CCCC1

#0 CC



#1 CCO



A1EKK · A2RT31;P62873

methyl (9R,10E,12E)-9-methoxyoctadeca-10,12-dienoate

family Olfactory receptor (vertebrate GPCR)
organism Mus musculus
PDB 9LKB
fold 1 sites 1 pocket 69.0 res
flavour (none) [llm abstain]
SMILES CCCCC/C=C/C=C/[C@@H](CCCCCCCC(=O)OC)OC

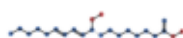
#0 CCCCC CO CO



#1 ? CO CO



#2 CO CO



#3 CCCCC CO



Catalogue

records 81–84 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1EKK · A2RT31

methyl (9R,10E,12E)-9-methoxyoctadeca-10,12-dienoate

family Olfactory receptor (vertebrate GPCR)
organism Mus musculus
PDB 9LKD
fold 1 sites 1 pocket 65.0 res
flavour (none) [llm_abstain]
SMILES CCCC/C=C/C=C/[C@@H](CCCCCCC(=O)OC)OC

#0 CCCCCC CO CO



#1 ? CO CO



#2 CO CO



#3 CCCCCC CO

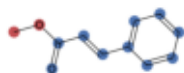


A1MBB · P59768;P62873;Q5FWY2;Q7TRW7

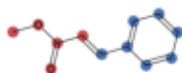
Methyl cinnamate

family Olfactory receptor (vertebrate GPCR)
organism Mus musculus
PDB 9W45
fold 4 sites 1 pocket 56.0 res
flavour fruity;sweet;woody [measured]
SMILES COC(=O)/C=C/c1ccccc1

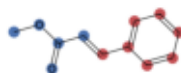
#0 CO



#1 CC(=O)OC



#2 Cc1ccccc1

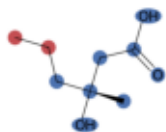


A1MBN · P59768;P62873

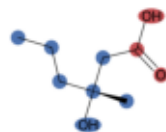
(3~{R})-3-methyl-3-oxidanyl-hexanoic acid

family Olfactory receptor (vertebrate GPCR)
organism Homo sapiens
PDB 9WG4
fold 1 sites 1 pocket 64.0 res
flavour (none) [llm_abstain]
SMILES CCC[C@@](C)(O)CC(=O)O

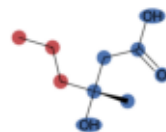
#0 CC



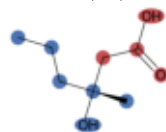
#1 O=CO



#2 CCC



#3 CC(=O)O

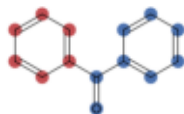


BZQ · P38405;P59768;P62873

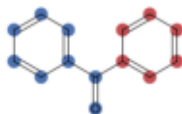
DIPHENYLMETHANONE

family Olfactory receptor (vertebrate GPCR)
organism 9LDX
fold 1 sites 1 pocket 59.0 res
flavour floral;fruity;medicinal;woody [measured]
SMILES O=C(c1ccccc1)c1ccccc1

#0 c1ccccc1



#1 c1ccccc1



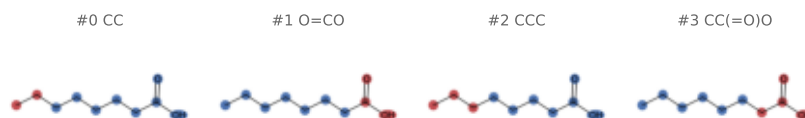
Catalogue

records 85–88 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

OCA · P59768;P62873;P63092

OCTANOIC ACID (CAPRYLIC ACID)

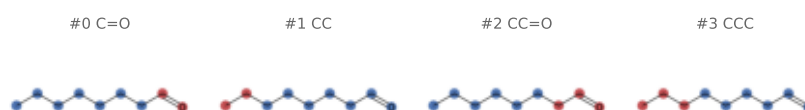
family Olfactory receptor (vertebrate GPCR)
organism Homo sapiens
PDB 8HTI
fold 4 sites 1 pocket 70.0 res
flavour dairy;fatty;green [measured]
SMILES CCCCCC(=O)O



OYA · P38405;P59768;P62873

OCTANAL

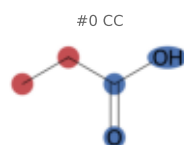
family Olfactory receptor (vertebrate GPCR)
organism Homo sapiens
PDB 9LE0
fold 1 sites 1 pocket 72.0 res
flavour citrus;fatty;green [measured]
SMILES CCCCCC=O



PPI · P59768;P62873;P63092;Q9H255

PROPANOIC ACID

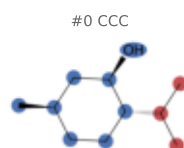
family Olfactory receptor (vertebrate GPCR)
organism Homo sapiens
PDB 8F76
fold 2 sites 1 pocket 51.0 res
flavour dairy;fatty;sour;spicy [measured]
SMILES CCC(=O)O



XUQ · P59768;P62873;P63092

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

family Olfactory receptor (vertebrate GPCR)
organism Homo sapiens
PDB 8UXY
fold 4 sites 1 pocket 58.0 res
flavour minty;roasted;woody [measured]
SMILES CC(C)[C@@H]1CC[C@H](C)C[C@H]1O



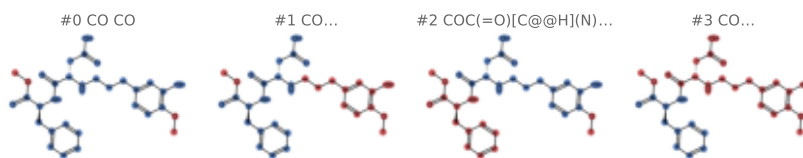
Catalogue

records 89–92 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1CD7 · Q8TE23;Q925D8

advantame

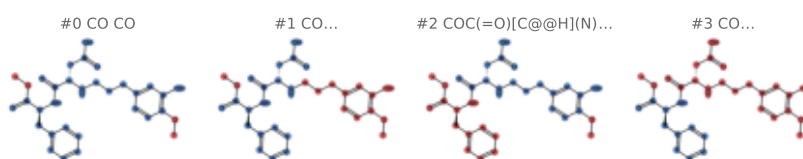
family T1R – sweet/umami receptor
organism Mus musculus
PDB 90Q2 90Q3
fold 1 sites 2 pocket 85.5 res
flavour sweet [llm]
SMILES COC(=O)[C@H](Cc1ccccc1)NC(=O)[C@H](CC(=O)O)C(=O)O



A1CD7 · Q7RTX0;Q8TE23

advantame

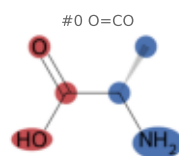
family T1R – sweet/umami receptor
organism Homo sapiens
PDB 90Q4 90Q5 90Q6
fold 1 sites 3 pocket 86.0 res
flavour sweet [llm]
SMILES COC(=O)[C@H](Cc1ccccc1)NC(=O)[C@H](CC(=O)O)C(=O)O



ALA · A0A173M094;A0A173M0G2

ALANINE

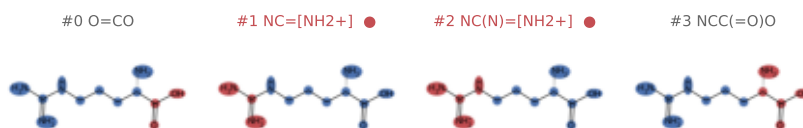
family T1R – sweet/umami receptor
organism Oryzias latipes
PDB 5X2N
fold 4 sites 4 pocket 51.2 res
flavour odorless [measured]
SMILES C[C@H](N)C(=O)O



ARG · A0A173M094;A0A173M0G2

ARGININE

family T1R – sweet/umami receptor
organism Oryzias latipes
PDB 5X2O
fold 4 sites 4 pocket 64.8 res
flavour odorless [measured]
SMILES NC(=[NH2+])NCCC[C@H](N)C(=O)O



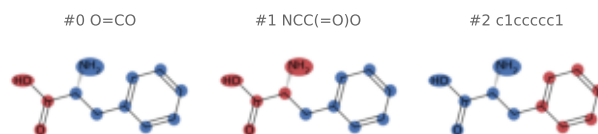
Catalogue

records 93–96 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

DPN · H2UNJ5;Q2MHK1

D-PHENYLALANINE

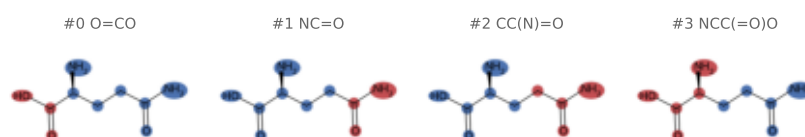
family T1R – sweet/umami receptor
organism Takifugu rubripes
PDB 9XPL
fold 2 sites 2 pocket 51.5 res
flavour sweet [measured]
SMILES N[C@H](Cc1ccccc1)C(=O)O



GLN · A0A173M094;A0A173M0G2

GLUTAMINE

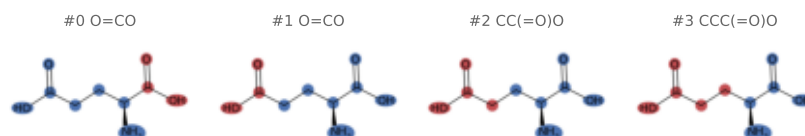
family T1R – sweet/umami receptor
organism Oryzias latipes
PDB 5X2M
fold 4 sites 4 pocket 63.5 res
flavour dairy;fatty;odorless;roasted [measured]
SMILES NC(=O)CC[C@H](N)C(=O)O



GLU · A0A173M094;A0A173M0G2

GLUTAMIC ACID

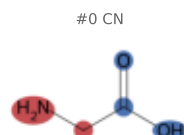
family T1R – sweet/umami receptor
organism Oryzias latipes
PDB 5X2P
fold 4 sites 4 pocket 63.5 res
flavour dairy;odorless;roasted [measured]
SMILES N[C@@H](CCC(=O)O)C(=O)O



GLY · A0A173M094;A0A173M0G2

GLYCINE

family T1R – sweet/umami receptor
organism Oryzias latipes
PDB 5X2Q
fold 4 sites 4 pocket 51.0 res
flavour fatty;odorless;sweet [measured]
SMILES NCC(=O)O



Catalogue

records 97-100 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1AEI · Q9NYV8

4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

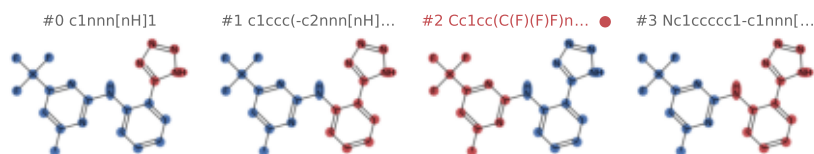
family T2R – bitter taste receptor
organism Homo sapiens
PDB 9IIW 9IJA
fold 3 sites 3 pocket 67.7 res
flavour (none) [llm_abstain]
SMILES Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n.



A1AEI · A8MTJ3;P59768;P62873;Q9NYV8

4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

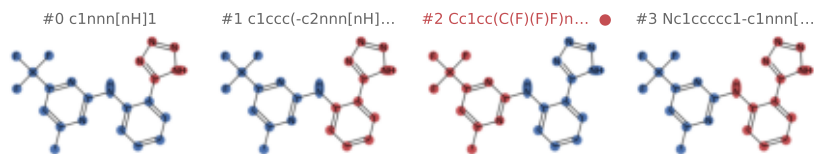
family T2R – bitter taste receptor
organism Homo sapiens
PDB 8VY9
fold 3 sites 1 pocket 76.0 res
flavour (none) [llm_abstain]
SMILES Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n.



A1AEI · P59768;P62873;P63096;Q9NYV8

4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

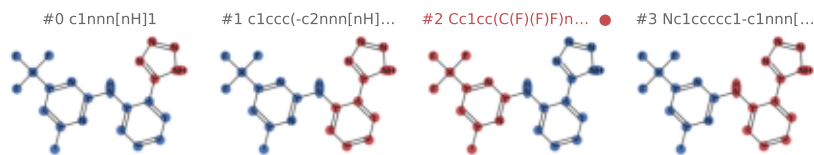
family T2R – bitter taste receptor
organism Homo sapiens
PDB 8VY7 9IJ9
fold 3 sites 2 pocket 76.5 res
flavour (none) [llm_abstain]
SMILES Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n.



A1AEI · P59768;P62873;Q9NYV8

4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

family T2R – bitter taste receptor
organism Homo sapiens
PDB 9IIX
fold 3 sites 2 pocket 79.5 res
flavour (none) [llm_abstain]
SMILES Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n.



Catalogue

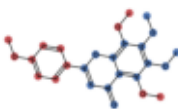
records 101–104 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1EUI · P59540;P59768;P62873;P63096

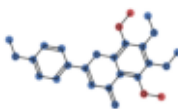
5,6,7,8-tetramethoxy-2-(4-methoxyphenyl)chromen-4-on...

family T2R – bitter taste receptor
organism Homo sapiens
PDB 9W0U
fold 3 sites 1 pocket 66.0 res
flavour bitter [measured]
SMILES COc1ccc(-c2cc(=O)c3c(OC)c(OC)c(OC)c(OC)c...

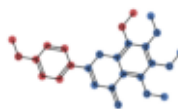
#0 COc1ccccc1 CO CO



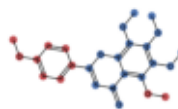
#1 CO CO



#2 COc1ccccc1 CO



#3 COc1ccccc1 CO

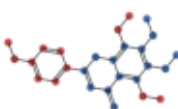


A1EUI · P59768;P62873;P63096;Q9NYV8

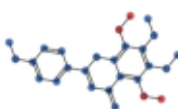
5,6,7,8-tetramethoxy-2-(4-methoxyphenyl)chromen-4-on...

family T2R – bitter taste receptor
organism Escherichia coli
PDB 9W0Q
fold 3 sites 1 pocket 70.0 res
flavour bitter [measured]
SMILES COc1ccc(-c2cc(=O)c3c(OC)c(OC)c(OC)c(OC)c...

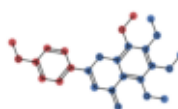
#0 COc1ccccc1 CO CO



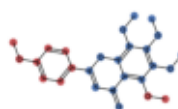
#1 CO CO



#2 COc1ccccc1 CO



#3 COc1ccccc1 CO

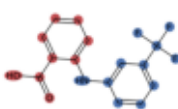


FLF · P59768;P62873;P63092;Q59310;Q9...

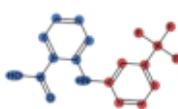
2-[[3-(TRIFLUOROMETHYL)PHENYL]AMINO] BENZOIC ACID

family T2R – bitter taste receptor
organism Homo sapiens
PDB 8XQR
fold 3 sites 1 pocket 73.0 res
flavour bitter [measured]
SMILES O=C(O)c1ccccc1Nc1ccc(C(F)(F)F)c1

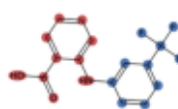
#0 O=C(O)c1ccccc1



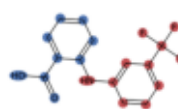
#1 FC(F)(F)c1ccccc1



#2 Nc1ccccc1C(=O)O



#3 Nc1ccc(C(F)(F)F)c1

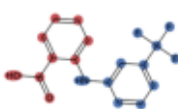


FLF · A8MTJ3;P59768;P62873;Q9NYV8

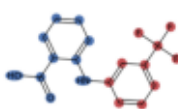
2-[[3-(TRIFLUOROMETHYL)PHENYL]AMINO] BENZOIC ACID

family T2R – bitter taste receptor
organism Homo sapiens
PDB 8RQL
fold 3 sites 2 pocket 70.5 res
flavour bitter [measured]
SMILES O=C(O)c1ccccc1Nc1ccc(C(F)(F)F)c1

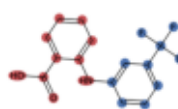
#0 O=C(O)c1ccccc1



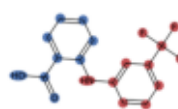
#1 FC(F)(F)c1ccccc1



#2 Nc1ccccc1C(=O)O



#3 Nc1ccc(C(F)(F)F)c1



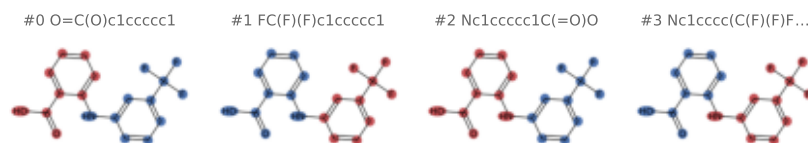
Catalogue

records 105–108 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

FLF · P59768;P62873;P63096;Q59310;Q9...

2-[[3-(TRIFLUOROMETHYL)PHENYL]AMINO] BENZOIC ACID

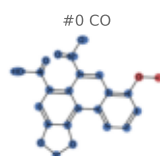
family T2R – bitter taste receptor
organism Homo sapiens
PDB 8XQS
fold 3 sites 1 pocket 70.0 res
flavour bitter [measured]
SMILES O=C(O)c1ccccc1Nc1ccc(C(F)(F)F)c1



GOQ · P59768;P62873;P63096;Q59310;Q9...

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-...

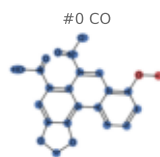
family T2R – bitter taste receptor
organism Homo sapiens
PDB 8XQN 8XQO
fold 3 sites 2 pocket 75.5 res
flavour bitter [measured]
SMILES C0c1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc1



GOQ · A8MTJ3;P59537;P59768;P62873

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-...

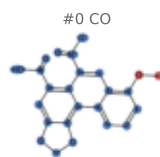
family T2R – bitter taste receptor
organism Homo sapiens
PDB 90XB
fold 3 sites 1 pocket 73.0 res
flavour bitter [measured]
SMILES C0c1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc1



GOQ · P0ABE7;P59537;P59768;P62873;P6...

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-...

family T2R – bitter taste receptor
organism Homo sapiens
PDB 90XA
fold 3 sites 1 pocket 75.0 res
flavour bitter [measured]
SMILES C0c1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc1



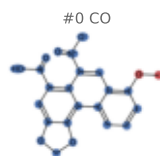
Catalogue

records 109–112 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

GOQ · A8MTJ3;P59768;P62873;Q59310;Q9...

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-...

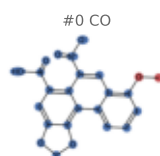
family T2R – bitter taste receptor
organism Homo sapiens
PDB 8XQP
fold 3 sites 1 pocket 73.0 res
flavour bitter [measured]
SMILES COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc.



GOQ · P59768;P62873;P63092;Q59310;Q9...

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-...

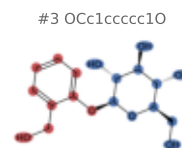
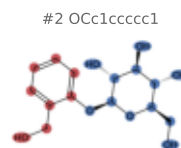
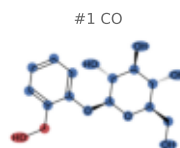
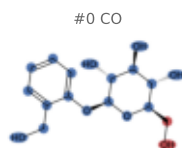
family T2R – bitter taste receptor
organism Homo sapiens
PDB 8XQL
fold 3 sites 2 pocket 74.0 res
flavour bitter [measured]
SMILES COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc.



SA0 · P04899;P59768;P62873;Q9NYV7

2-(hydroxymethyl)phenyl beta-D-glucopyranoside

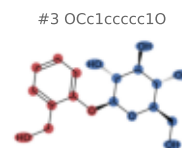
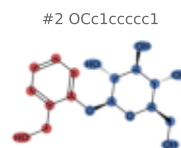
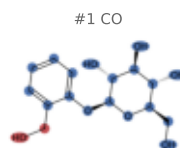
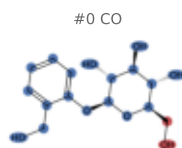
family T2R – bitter taste receptor
organism Homo sapiens
PDB 9K6L
fold 2 sites 1 pocket 70.0 res
flavour bitter [measured]
SMILES OCc1cccc1O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C@@H]1O



SA0 · P59768;P62873;P63092;Q9NYV7

2-(hydroxymethyl)phenyl beta-D-glucopyranoside

family T2R – bitter taste receptor
organism Homo sapiens
PDB 9KPD
fold 2 sites 1 pocket 68.0 res
flavour bitter [measured]
SMILES OCc1cccc1O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C@@H]1O



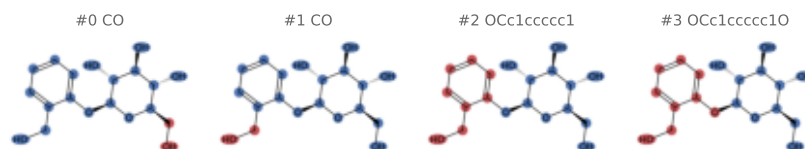
Catalogue

records 113–116 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

SA0 · P59768;P62873;P63096;Q9NYV7

2-(hydroxymethyl)phenyl beta-D-glucopyranoside

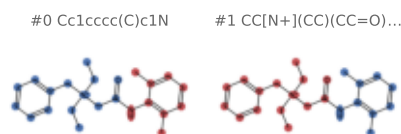
family T2R – bitter taste receptor
organism Homo sapiens
PDB 9KPE 9KPF
fold 2 sites 2 pocket 68.0 res
flavour bitter [measured]
SMILES OCc1ccccc1O[C@@H]1O[C@H](CO)[C@@H](O)[C@H]1O



WK3 · P59540;P59768;P62873;P63096

N-benzyl-2-(2,6-dimethylanilino)-N,N-diethyl-2-oxoet...

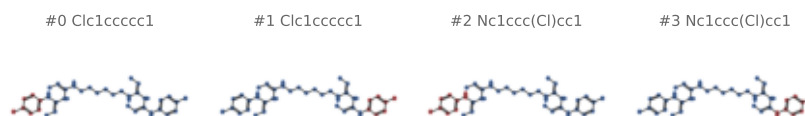
family T2R – bitter taste receptor
organism Homo sapiens
PDB 9W0T
fold 3 sites 1 pocket 57.0 res
flavour (none) [llm_abstain]
SMILES CC[N+](CC)(CC(=O)Nc1c(C)cccc1C)Cc1ccccc1



XC9 · P0ABE7;P59768;P62873;P63096;Q9...

1-[6-[azanylidene-[azanylidene-[(4-chlorophenyl)am...

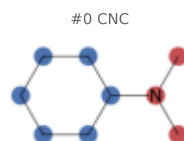
family T2R – bitter taste receptor
organism Escherichia coli
PDB 9W0P
fold 4 sites 1 pocket 69.0 res
flavour bitter [measured]
SMILES [H]/N=C(/NCCCCCN/C(=N\[H])N/C(=N\[H])Nc...



8IA · P59768;P62873;P63092;Q5QD08

~{N},~{N}-dimethylcyclohexanamine

family TAAR – olfactory amine receptor..
organism Mus musculus
PDB 8PM2
fold 1 sites 1 pocket 54.0 res
flavour (none) [llm_abstain]
SMILES CN(C)C1CCCC1



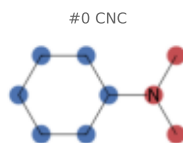
Catalogue

records 117–120 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

8IA · P59768;P62873;Q5QD04

~{N},~{N}-dimethylcyclohexanamine

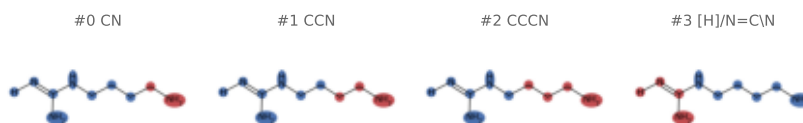
family TAAR — olfactory amine receptor...
organism Mus musculus
PDB 8ITF
fold 1 sites 1 pocket 54.0 res
flavour (none) [llm_abstain]
SMILES CN(C)C1CCCC1



AG2 · P59768;P62873;Q5QNQ1

AGMATINE

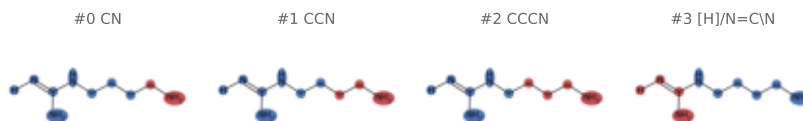
family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M0X
fold 4 sites 2 pocket 59.0 res
flavour bitter [mined]
SMILES [H]/N=C(\N)NCCCCN



AG2 · P59768;P62873;P63092;Q5QNP8

AGMATINE

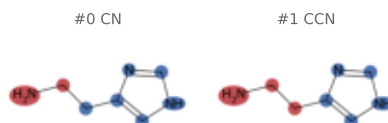
family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M40
fold 4 sites 1 pocket 64.0 res
flavour bitter [mined]
SMILES [H]/N=C(\N)NCCCCN



HSM · P59768;P62873;Q5QNP9

HISTAMINE

family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M3S
fold 4 sites 1 pocket 62.0 res
flavour bitter [mined]
SMILES NCCc1c[nH]cn1



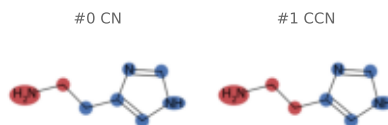
Catalogue

records 121–124 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

HSM · P59768;P62873;Q5QNQ1

HISTAMINE

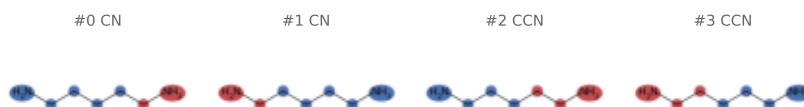
family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M1P
fold 4 sites 1 pocket 62.0 res
flavour bitter [mined]
SMILES NCCc1c[nH]cn1



N2P · P59768;P62873;Q5QNP2

PENTANE-1,5-DIAMINE

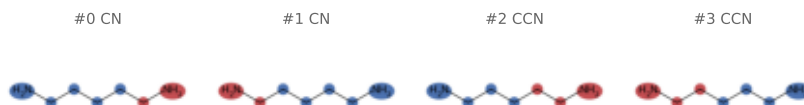
family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M2S
fold 4 sites 1 pocket 59.0 res
flavour bitter [mined]
SMILES NCCCCN



N2P · P59768;P62873;Q5QNQ1

PENTANE-1,5-DIAMINE

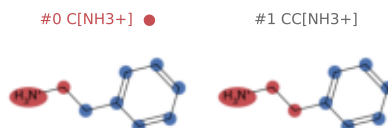
family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M0Z
fold 4 sites 1 pocket 60.0 res
flavour bitter [mined]
SMILES NCCCCN



PEA · P59768;P62873;P63092;P63096;Q5...

2-PHENYLETHYLAMINE

family TAAR — olfactory amine receptor...
organism Mus musculus
PDB 8IW7
fold 4 sites 1 pocket 60.0 res
flavour meaty;medicinal [measured]
SMILES [NH3+][Cc1ccccc1]



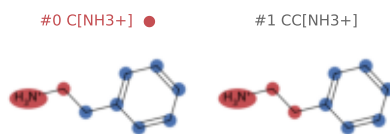
Catalogue

records 125–128 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PEA · Q5QD04

2-PHENYLETHYLAMINE

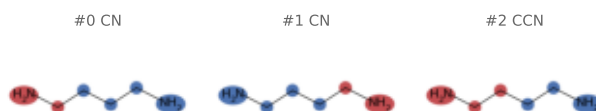
family TAAR — olfactory amine receptor...
organism Mus musculus
PDB 8IWM
fold 4 sites 1 pocket 60.0 res
flavour meaty;medicinal [measured]
SMILES [NH3+][C]Cc1ccccc1



PUT · P59768;P62873;Q5QNQ1

1,4-DIAMINOBTANE

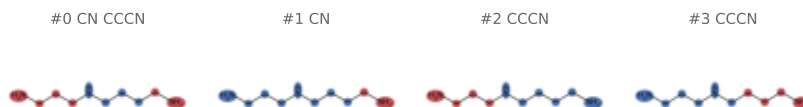
family TAAR — olfactory amine receptor...
organism Danio rerio
PDB 9M20
fold 4 sites 1 pocket 61.0 res
flavour (none) [llm_abstain]
SMILES NCCCCN



SPD · Q5QD04

SPERMIDINE

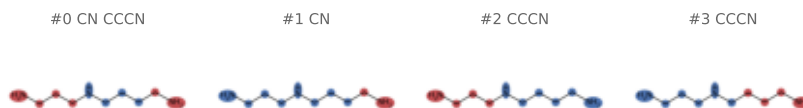
family TAAR — olfactory amine receptor...
organism Mus musculus
PDB 8IWE
fold 4 sites 1 pocket 61.0 res
flavour (none) [llm_abstain]
SMILES NCCCCNCCCN



SPD · P59768;P62873;P63092;P63096;Q5 ...

SPERMIDINE

family TAAR — olfactory amine receptor...
organism Mus musculus
PDB 8IW4
fold 4 sites 1 pocket 61.0 res
flavour (none) [llm_abstain]
SMILES NCCCCNCCCN



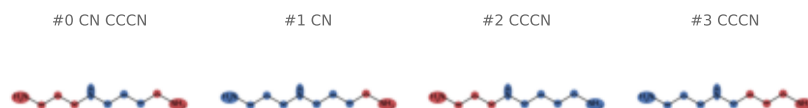
Catalogue

records 129–132 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

SPD · P59768;P62873;Q5QNP2

SPERMIDINE

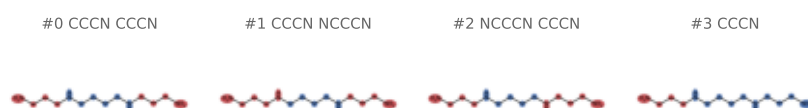
family TAAR — olfactory amine receptor...
organism *Danio rerio*
PDB 9M3Q
fold 4 sites 1 pocket 71.0 res
flavour (none) [llm_abstain]
SMILES NCCCCNCCCN



SPM · A0A590UJY2;A0A678XLR1;P59768;P...

SPERMINE

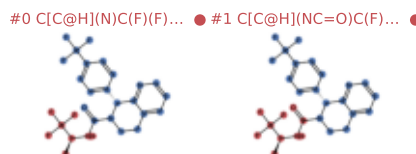
family TAAR — olfactory amine receptor...
organism *Petromyzon marinus*
PDB 9VMG
fold 2 sites 1 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES NCCCNCCCCNCCCN



A1AIY · Q8R4D5

(8R)-8-[4-(trifluoromethyl)phenyl]-N-[(2S)-1,1,1-tri...

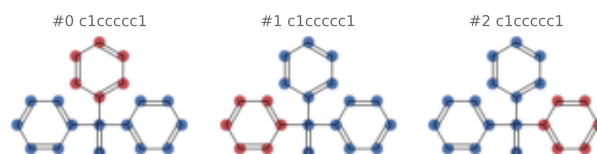
family TRP taste/chemesthesis channel
organism *Mus musculus*
PDB 9B6F
fold 4 sites 4 pocket 72.8 res
flavour (none) [llm_abstain]
SMILES C[C@H](NC(=O)N1CCc2ccnc2[C@H]1c1ccc(C(F)(F)F)c1)C(F)(F)F



A1CG5 · S5UH55

oxotri(phenyl)-lambda~5~-phosphane

family TRP taste/chemesthesis channel
organism *Danio rerio*
PDB 9P3T
fold 3 sites 4 pocket 77.0 res
flavour (none) [llm_abstain]
SMILES O=P(c1ccccc1)(c1ccccc1)c1ccccc1



Catalogue

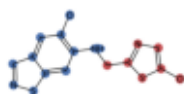
records 133–136 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1CGW · A0A5P9VSM8;S5UH55

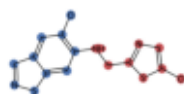
5-chloro-N-[(5-chloro-1,3-thiazol-2-yl)methyl]-1,2-b...

family TRP taste/chemesthesis channel
organism Danio rerio
PDB 9P3U 9P3V
fold 3 sites 8 pocket 71.5 res
flavour (none) [llm_abstain]
SMILES Clc1cnc(CNc2cc3sncc3cc2Cl)s1

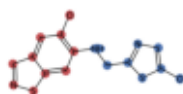
#0 Cc1ncc(Cl)s1 ●



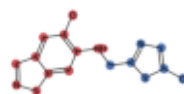
#1 NCc1ncc(Cl)s1 ●



#2 Clc1ccc2sncc2c1 ●



#3 Nc1cc2sncc2cc1Cl ●

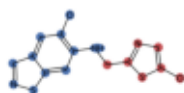


A1CGW · S5UH55

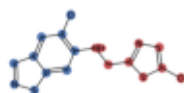
5-chloro-N-[(5-chloro-1,3-thiazol-2-yl)methyl]-1,2-b...

family TRP taste/chemesthesis channel
organism Danio rerio
PDB 9P3N 9P3O 9P3P
fold 3 sites 12 pocket 69.7 res
flavour (none) [llm_abstain]
SMILES Clc1cnc(CNc2cc3sncc3cc2Cl)s1

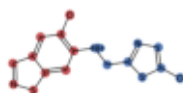
#0 Cc1ncc(Cl)s1 ●



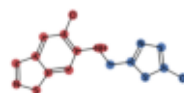
#1 NCc1ncc(Cl)s1 ●



#2 Clc1ccc2sncc2c1 ●



#3 Nc1cc2sncc2cc1Cl ●

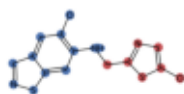


A1CGW · A0AB32TV16

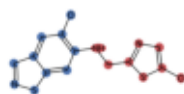
5-chloro-N-[(5-chloro-1,3-thiazol-2-yl)methyl]-1,2-b...

family TRP taste/chemesthesis channel
organism Danio rerio
PDB 9P3S
fold 3 sites 4 pocket 68.0 res
flavour (none) [llm_abstain]
SMILES Clc1cnc(CNc2cc3sncc3cc2Cl)s1

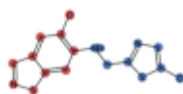
#0 Cc1ncc(Cl)s1 ●



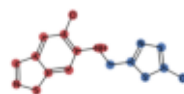
#1 NCc1ncc(Cl)s1 ●



#2 Clc1ccc2sncc2c1 ●



#3 Nc1cc2sncc2cc1Cl ●

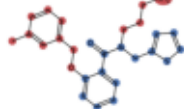


LQ7 · Q8R4D5

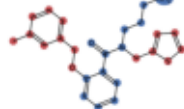
N-(3-aminopropyl)-2-[(3-methylphenyl)methoxy]-N-[(th...

family TRP taste/chemesthesis channel
organism Mus musculus
PDB 9B6G
fold 4 sites 4 pocket 87.0 res
flavour (none) [llm_abstain]
SMILES Cc1cccc(COc2ccccc2C(=O)N(CCCN)Cc2cccs2)c...

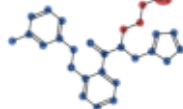
#0 Cc1cccc(CO)c1...



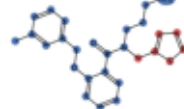
#1 Cc1cccc(CO)c1...



#2 CCCN



#3 Cc1cccs1



Catalogue

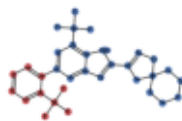
records 137–140 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

T14 · A0A5S8WF66

3-{7-(trifluoromethyl)-5-[2-(trifluoromethyl)phenyl]}...

family TRP taste/chemesthesis channel
organism Parus major
PDB 6072 9B6I
fold 4 sites 8 pocket 88.5 res
flavour (none) [llm_abstain]
SMILES FC(F)(F)c1ccccc1-c1cc(C(F)(F)F)c2[nH]c(C...

#0 FC(F)(F)c1ccccc1

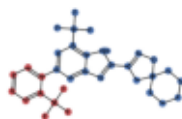


T14 · Q8R4D5

3-{7-(trifluoromethyl)-5-[2-(trifluoromethyl)phenyl]}...

family TRP taste/chemesthesis channel
organism Mus musculus
PDB 9B6E 9B6H
fold 4 sites 8 pocket 89.8 res
flavour (none) [llm_abstain]
SMILES FC(F)(F)c1ccccc1-c1cc(C(F)(F)F)c2[nH]c(C...

#0 FC(F)(F)c1ccccc1



ULO · Q8R4D5

nonyl(oxo)di(propan-2-yl)-lambda~5~-phosphane

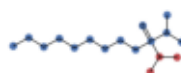
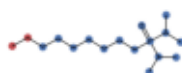
family TRP taste/chemesthesis channel
organism Mus musculus
PDB 8E4L 8E4M 9B6H
fold 4 sites 12 pocket 81.7 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCP(=O)(C(C)C)C(C)C

#0 CC

#1 CCC

#2 CCC

#3 CCC



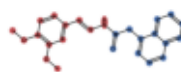
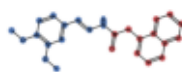
YUS · S5UH55

N'-[(E)-(3,4-dimethoxyphenyl)methylidene]-2-(naphtha...

family TRP taste/chemesthesis channel
organism Danio rerio
PDB 7MBV
fold 3 sites 4 pocket 62.0 res
flavour (none) [llm_abstain]
SMILES C0c1ccc(/C=N/NC(=O)Cc2cccc3ccccc23)cc1OC

#0 O=CCc1cccc2ccccc...

#1 COc1ccc(/C=N/N)c...



Catalogue

records 141–144 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

OIG · O75762

~{N}-[4-[4-[(azanylidene-\$l^{4}\$-azanylidene)amino]ph...

family TRPA1 – irritant/pungency receptor...

organism Homo sapiens

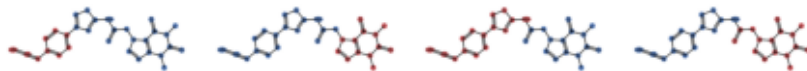
PDB 70R0 70R1

fold 0 sites 8 pocket 58.1 res

flavour (none) [llm_abstain]

SMILES Cn1c(=O)c2c(ncn2CC(=O)Nc2nc(-c3ccc(N=[N+]

#0 N=[N+]=Nc1ccccc1 ● #1 Cn1c(=O)c2ncnc2n... #2 N=[N+]=Nc1ccc(-c... ● #3 Cn1c(=O)c2c(ncn2...



6OU · O75762

[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3...

family TRPA1 – irritant/pungency receptor...

organism Homo sapiens

PDB 6PQ0 6PQP 6PQQ

fold 0 sites 56 pocket 43.9 res

flavour (none) [llm_abstain]

SMILES CCCCCC/C=C\CCCCCCC(=O)O[C@H](COC(=O)C...

#0 CCCCCCCCC... #1 CCCCCCCCC... #2 CCCCCCCCC... #3 CCCCCCCCC...



9BE · O75762

N-benzylthioformamide

family TRPA1 – irritant/pungency receptor...

organism Homo sapiens

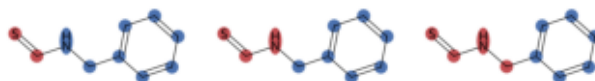
PDB 6PQP

fold 0 sites 4 pocket 50.0 res

flavour (none) [llm_abstain]

SMILES S=CNCc1ccccc1

#0 C=S #1 NC=S #2 CNC=S



A1BNO · O75762

(2S)-2-{3-methyl-1-[(3-methyl-1,2,4-oxadiazol-5-yl)m...

family TRPA1 – irritant/pungency receptor...

organism Homo sapiens

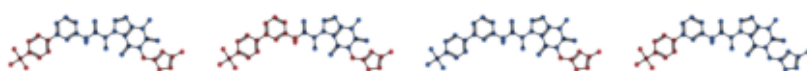
PDB 9MOE

fold 0 sites 4 pocket 66.8 res

flavour (none) [llm_abstain]

SMILES Cc1noc(Cn2c(=O)c3c(ncn3[C@@H](C)C(=O)Nc3...

#0 Cc1noc(C)n1... ● #1 Cc1noc(C)n1... ● #2 Cc1noc(C)n1 #3 FC(F)(F)c1ncccn1 ●



Catalogue

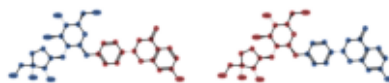
records 145–148 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1ETS · 075762

Liquiritin apioside

family TRPA1 – irritant/pungency receptor..
organism Homo sapiens
PDB 9VU1
fold 0 sites 4 pocket 55.2 res
flavour (none) [llm_abstain]
SMILES O=C1C[C@@H](c2ccc(O[C@@H]3O[C@H](CO)[C@@H](O)[C@H]3O)c2)O

#0 O=C1C[C@@H](c2ccc... #1 OC[C@H]1O[C@@H](...



JT0 · 075762

2-chloro-N-[4-(4-methoxyphenyl)-1,3-thiazol-2-yl]-N...

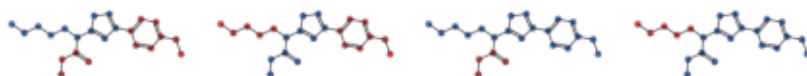
family TRPA1 – irritant/pungency receptor..
organism Homo sapiens
PDB 6PQ0
fold 0 sites 4 pocket 48.0 res
flavour (none) [llm_abstain]
SMILES COCCCN(C(=O)C(=O)Cl)c1nc(-c2ccc(OC)cc2)cs1

#0 O=CCCl...

#1 CCCOC COc1cccc1

#2 O=CCCl

#3 CCCOC



LXY · 075762

(4R,5S)-4-fluoro-1-[(4-fluorophenyl)sulfonyl]-5-meth...

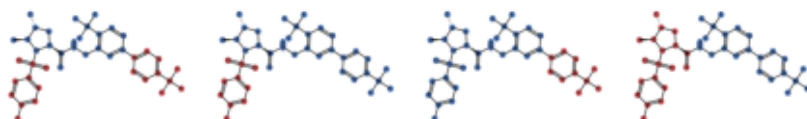
family TRPA1 – irritant/pungency receptor..
organism Homo sapiens
PDB 6WJ5
fold 0 sites 4 pocket 90.8 res
flavour (none) [llm_abstain]
SMILES C[C@H]1[C@H](F)C[C@H](C(=O)NCc2cc(-c3cn...

#0 O=S(=O)c1ccc(F)c... ●

#1 O=S(=O)c1ccc(F)c...

#2 FC(F)(F)c1ncccn1 ●

#3 C[C@H]1[C@H](F)C... ●



ULJ · 075762

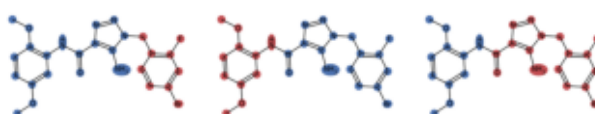
5-amino-1-[(4-bromo-2-fluorophenyl)methyl]-N-(2,5-di...

family TRPA1 – irritant/pungency receptor..
organism Homo sapiens
PDB 6X2J
fold 0 sites 4 pocket 81.2 res
flavour (none) [llm_abstain]
SMILES COc1ccc(OC)c(NC(=O)c2nnn(Cc3ccc(Br)cc3F)...

#0 Cc1ccc(Br)cc1F ●

#1 COc1ccc(OC)c(N)c...

#2 Nc1c(C=O)nnn1Cc1... ●



Catalogue

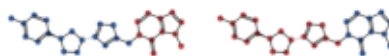
records 149–152 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

VKM · O75762

1-({3-[(3R,5R)-5-(4-fluorophenyl)oxolan-3-yl]-1,2,4-...

family TRPA1 – irritant/pungency receptor..
organism Homo sapiens
PDB 7JUP
fold 0 sites 4 pocket 75.0 res
flavour (none) [llm_abstain]
SMILES Cn1cnc2ncn(Cc3nc([C@@H]4C0[C@@H](c5ccc(F...

#0 Cn1cnc2ncnc(=O)c... ● #1 Cc1nc([C@@H]2CO[... ●

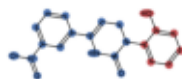


KX7 · noUP:6NR3

Icilin

family TRPM8 – cooling/menthol receptor..
organism Ficedula albicollis
PDB 6NR3
fold 4 sites 4 pocket 72.0 res
flavour minty [llm]
SMILES O=C1NC(c2cccc([N+](=O)[O-])c2)=CCN1c1ccc...

#0 Oc1cccc1

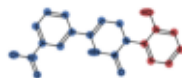


KX7 · Q8R4D5

Icilin

family TRPM8 – cooling/menthol receptor..
organism Mus musculus
PDB 7WRC 7WRD 7WRE 7WRF 9VJR 9VNC
fold 4 sites 24 pocket 69.8 res
flavour minty [llm]
SMILES O=C1NC(c2cccc([N+](=O)[O-])c2)=CCN1c1ccc...

#0 Oc1cccc1



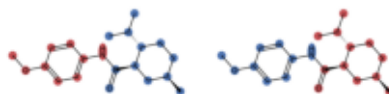
KXS · noUP:6NR2

(1R,2S,5R)-N-(4-methoxyphenyl)-5-methyl-2-(propan-2-...

family TRPM8 – cooling/menthol receptor..
organism Ficedula albicollis
PDB 6NR2
fold 3 sites 4 pocket 79.0 res
flavour minty [llm]
SMILES COc1ccc(NC(=O)[C@@H]2C[C@H](C)CC[C@H]2C[...]

#0 COc1ccc(N)cc1

#1 CC(C)[C@@H]1CC[C...



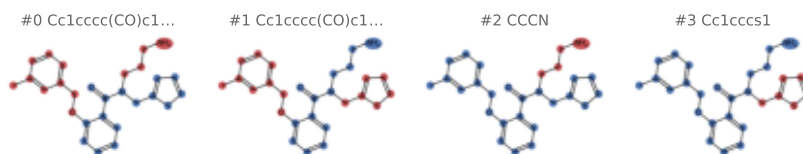
Catalogue

records 153–156 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

LQ7 · A0A5S8WF66

N-(3-aminopropyl)-2-[(3-methylphenyl)methoxy]-N-[(th...

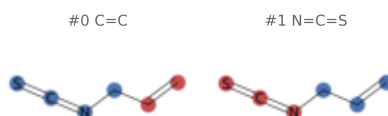
family TRPM8 — cooling/menthol receptor...
organism Parus major
PDB 606R
fold 4 sites 4 pocket 82.0 res
flavour (none) [llm_abstain]
SMILES Cc1cccc(C0c2ccccc2C(=O)N(CCCN)Cc2cccs2)C...



UL9 · Q8R4D5

3-isothiocyantoprop-1-ene

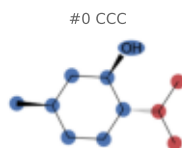
family TRPM8 — cooling/menthol receptor...
organism Mus musculus
PDB 8E4L
fold 4 sites 4 pocket 46.0 res
flavour bitter;spicy;sulfurous [measured]
SMILES C=CCN=C=S



XUQ · A0A5S8WF66

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

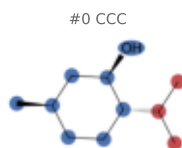
family TRPM8 — cooling/menthol receptor...
organism Parus major
PDB 9PB6 9ZCQ 9ZEZ
fold 4 sites 12 pocket 68.0 res
flavour minty;roasted;woody [measured]
SMILES CC(C)[C@@H]1CC[C@@H](C)C[C@H]1O



XUQ · Q7Z2W7

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

family TRPM8 — cooling/menthol receptor...
organism Homo sapiens
PDB 9PB5
fold 4 sites 4 pocket 60.0 res
flavour minty;roasted;woody [measured]
SMILES CC(C)[C@@H]1CC[C@@H](C)C[C@H]1O



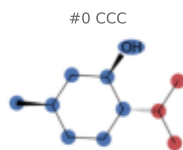
Catalogue

records 157–160 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XUQ · Q8R4D5

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

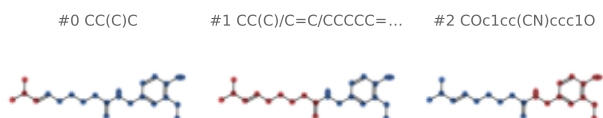
```
family      TRPM8 - cooling/menthol receptor...
organism    Mus musculus
PDB         9VJQ
fold 4      sites 4      pocket 53.0 res
flavour     minty;roasted;woody [measured]
SMILES      CC(C)[C@H]1CC[C@H](C)C[C@H]1O
```



4DY · 035433

(6E)-N-(4-hydroxy-3-methoxybenzyl)-8-methylnon-6-ena...

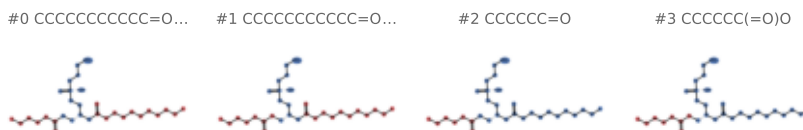
family	TRPV1 – capsaicin/vanilloid receptor...
organism	Rattus norvegicus
PDB	7LPA 7LPB 7LPD 7LPE
fold 0	sites 16 pocket 78.5 res
flavour	herbal;odorless [measured]
SMILES	C0c1cc(CNC(=O)CCCC/C=C/C(C)C)ccc1O



651 · 035433;P0CH43

(9R,12R)-15-amino-12-hydroxy-6,12-dioxo-7,11,13-trio...

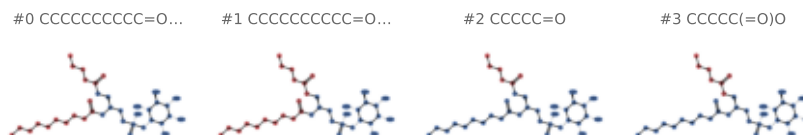
```
family      TRPV1 - capsaicin/vanilloid receptor...
organism    Rattus norvegicus
PDB         7L2R 7L2T
fold 0      sites 8   pocket 54.0 res
flavour     (none) [llm abstain]
SMILES      CCCCCCCCCC(=O)OC(=O)C(C(=O)CCCC)COP(=O)(O)O
```



6ES · 035433

(2S)-1-{[(R)-hydroxy{[(1R,2R,3S,4S,5S,6S)-2,3,4,5,6-...

```
family      TRPV1 - capsaicin/vanilloid receptor...
organism    Rattus norvegicus
PDB         5IRZ
fold 0      sites 4   pocket 97.0 res
flavour     (none) [llm abstain]
SMILES      CCCCCCCCCC(=O)O[C@@H](COC(=O)CCCC)COP(=O)(O)O
```



Catalogue

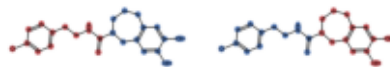
records 161–164 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

6ET · O35433

capsazepine

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 5IS0
fold 0 sites 4 pocket 79.0 res
flavour (none) [llm_abstain]
SMILES Oc1cc2c(cc1O)CN(C(=S)NCCc1ccc(Cl)cc1)CCG.

#0 S=CNCCc1ccc(Cl)c... #1 Oc1cc2c(cc1O)CNC...

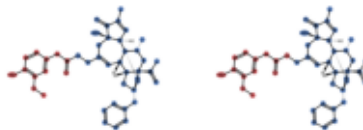


6EU · O35433

resiniferatoxin

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 7L2L 7L2N 7L2O 7L2V 7L2W 7L2X 7MZ5 7MZ7..
fold 0 sites 72 pocket 94.1 res
flavour spicy [llm]
SMILES C=C(C)[C@]12C[C@@H](C)[C@@]340[C@](Cc5cc...

#0 COc1cc(CC=O)ccc1... #1 COc1cc(CC(=O)O)c...

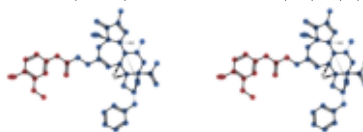


6EU · O35433;POCH43

resiniferatoxin

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 5IRX 7L2M
fold 0 sites 8 pocket 86.9 res
flavour spicy [llm]
SMILES C=C(C)[C@]12C[C@@H](C)[C@@]340[C@](Cc5cc...

#0 COc1cc(CC=O)ccc1... #1 COc1cc(CC(=O)O)c...

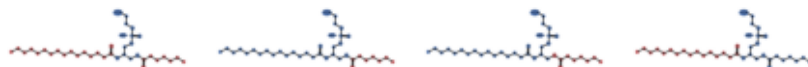


6IY · O35433

(10R,13S)-16-amino-13-hydroxy-7,13-dioxo-8,12,14-tri...

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 7L2P 7L2S 7L2W 7L2X 7MZ6
fold 0 sites 20 pocket 59.8 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCCCCCCCC(=O)O[C@H](COC(=O)CCCCC...

#0 CCCCCCCCCCCCCCCC... #1 CCCCCCCC=O #2 CCCCCCCC(=O)O #3 CCCCCCCCCCCCCCCC...



Catalogue

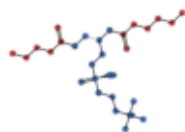
records 165–168 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

608 · 035433;POCH43

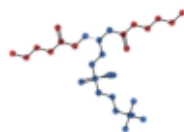
(4R,7S)-4-hydroxy-N,N,N-trimethyl-4,9-dioxo-7-[(pent...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 5IRX
fold 0 sites 4 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES CCCCC(=O)O[C@@H](COC(=O)CCCC)COP(=O)(O)C

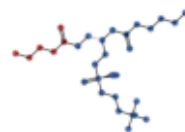
#0 CCCCC=O CCCCC=O



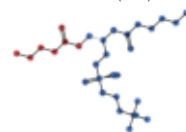
#1 CCCCC=O...



#2 CCCCC=O



#3 CCCCC(=O)O

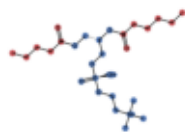


608 · 035433

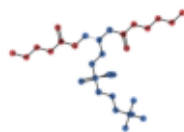
(4R,7S)-4-hydroxy-N,N,N-trimethyl-4,9-dioxo-7-[(pent...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 5IRZ
fold 0 sites 4 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES CCCCC(=O)O[C@@H](COC(=O)CCCC)COP(=O)(O)C

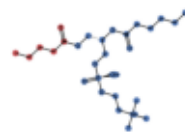
#0 CCCCC=O CCCCC=O



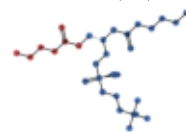
#1 CCCCC=O...



#2 CCCCC=O



#3 CCCCC(=O)O

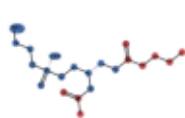


609 · 035433;POCH43

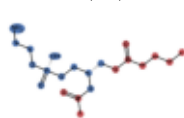
(2S)-2-(acetyloxy)-3-[[R)-(2-aminoethoxy)(hydroxy)p...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 5IRX
fold 0 sites 4 pocket 36.2 res
flavour (none) [llm_abstain]
SMILES CCCC(=O)OC[C@@H](COP(=O)(O)OCCN)OC(C)C

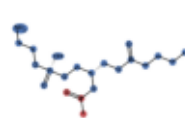
#0 CCCCC=O CC=O



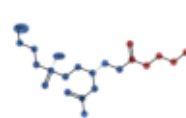
#1 CCCCC(=O)O CC=O



#2 CC=O



#3 CCCCC=O

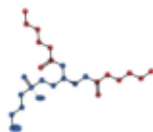


60E · 035433;POCH43

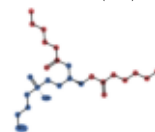
(2S)-3-[[S)-(2-aminoethoxy)(hydroxy)phosphoryl]oxy]...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 5IRX
fold 0 sites 8 pocket 41.5 res
flavour (none) [llm_abstain]
SMILES CCCCC(=O)OC[C@@H](COP(=O)(O)OCCN)OC(C)C

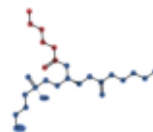
#0 CCCCC=O...



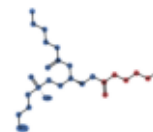
#1 CCCCC(=O)O...



#2 CCCCC=O



#3 CCCCC=O



Catalogue

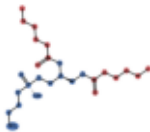
records 169–172 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

60E · O35433

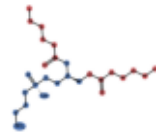
(2S)-3-[[[(S)-(2-aminoethoxy)(hydroxy)phosphoryl]oxy]...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 5IRZ
fold 0 sites 16 pocket 38.2 res
flavour (none) [llm_abstain]
SMILES CCCCC(=O)OC[C@H](COP(=O)(O)OCCN)OC..

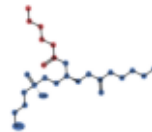
#0 CCCCCC=O...



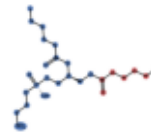
#1 CCCCCC(=O)O...



#2 CCCCCC=O



#3 CCCCCC=O

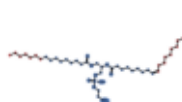


60U · Q8NER1

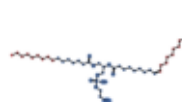
[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 8YCP
fold 0 sites 4 pocket 67.5 res
flavour (none) [llm_abstain]
SMILES CCCCCCC/C=C\CCCCCCC(=O)O[C@H](COC(=O)C..

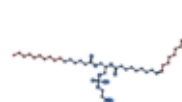
#0 CCCCCCCCC...



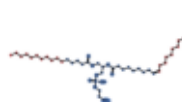
#1 CCCCCCCCC...



#2 CCCCCCCCC...



#3 CCCCCCCCC...

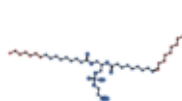


60U · O35433

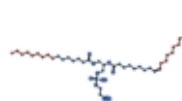
[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 7LP9 7LPA 7LPB 7LPC 7LPD 7LPE 7RQU 7RQV..
fold 0 sites 160 pocket 46.0 res
flavour (none) [llm_abstain]
SMILES CCCCCCC/C=C\CCCCCCC(=O)O[C@H](COC(=O)C..

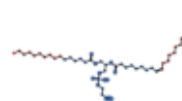
#0 CCCCCCCCC...



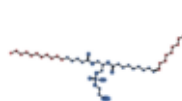
#1 CCCCCCCCC...



#2 CCCCCCCCC...



#3 CCCCCCCCC...

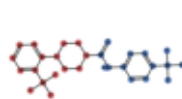
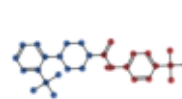
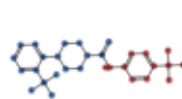
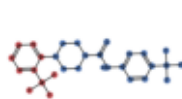


A1C8H · Q8NER1

4-[3-(trifluoromethyl)pyridin-2-yl]-N-[5-(trifluorom...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 1IC0
fold 0 sites 4 pocket 76.0 res
flavour (none) [llm_abstain]
SMILES O=C(Nc1ccc(C(F)(F)F)cn1)N1CCN(c2ncccc2C...

#0 FC(F)(F)c1ccnc1 ● #1 Nc1ccc(C(F)(F)F)... ● #2 O=CNc1ccc(C(F)(F)F)... ● #3 FC(F)(F)c1ccnc1... ●



Catalogue

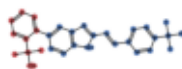
records 173–176 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1C8I · Q8NER1

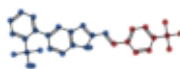
2-[(2P)-2-(2-[(E)-2-[4-(trifluoromethyl)phenyl]ethen...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 11CN
fold 0 sites 4 pocket 84.0 res
flavour (none) [llm_abstain]
SMILES CC(C)(O)c1ccccc1-c1ccc2[nH]c(/C=C/c3ccc(

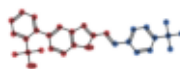
#0 CC(C)(O)c1ccccc1



#1 Cc1ccc(C(F)(F)F)...



#2 Cc1nc2cc(-c3cccc... ●

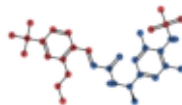


A1C8J · Q8NER1

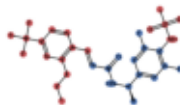
(2E)-N-[(1R)-1-[3,5-difluoro-4-(methanesulfonamido)p...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 11CL
fold 0 sites 4 pocket 75.0 res
flavour (none) [llm_abstain]
SMILES CCCC1nc(C(F)(F)F)ccc1/C=C/C(=O)N[C@H](C)...

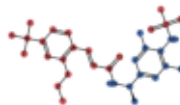
#0 Cc1ccc(C(F)(F)F)... ●



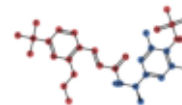
#1 Cc1ccc(C(F)(F)F)... ●



#2 CCCC1nc(C(F)(F)F)... ●



#3 CCCC1nc(C(F)(F)F)... ●

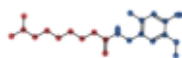


A1C8K · Q8NER1

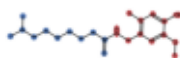
N-[(4-hydroxy-2-iodo-5-methoxyphenyl)methyl]-8-methy...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 11CK
fold 0 sites 4 pocket 80.0 res
flavour (none) [llm_abstain]
SMILES COc1cc(CNC(=O)CCCCCCC(C)C)c(I)cc1O

#0 CC(C)CCCCCCC=O



#1 COc1cc(CN)c(I)cc... ●

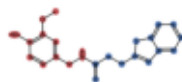


A1CHV · Q8NER1

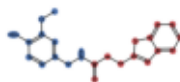
3-(1-benzothiophen-2-yl)-N-[(4-hydroxy-3-methoxyphen...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 9PEA
fold 0 sites 4 pocket 78.0 res
flavour (none) [llm_abstain]
SMILES COc1cc(CNC(=O)CCc2cc3ccccc3s2)ccc1O

#0 COc1cc(CN)ccc1O



#1 O=CCc1cc2ccccc2... ●



Catalogue

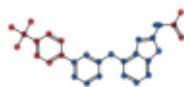
records 177–180 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1D6R · O35433

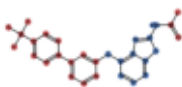
~{N}-[4-[6-[4-(trifluoromethyl)phenyl]pyrimidin-4-yl...

family TRPV1 — capsaicin/vanilloid receptor...
organism Escherichia coli K-12
PDB 9W4M
fold 0 sites 4 pocket 80.2 res
flavour (none) [llm_abstain]
SMILES CC(=O)Nc1nc2c(oc3cc(-c4ccc(C(F)(F)F)cc4)...

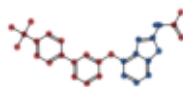
#0 CC=O...



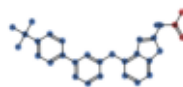
#1 CC=O... ●



#2 CC=O... ●



#3 CC=O

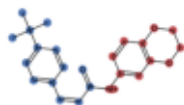


A1D6W · O35433

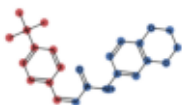
(~{Z})-3-(4-~{tert}-butylphenyl)-~{N}-(2,3-dihydro-1...

family TRPV1 — capsaicin/vanilloid receptor...
organism Escherichia coli K-12
PDB 9W4T
fold 0 sites 4 pocket 84.0 res
flavour (none) [llm_abstain]
SMILES CC(C)(C)c1ccc(/C=C\C(=O)Nc2ccc3c(c2)OCCO...

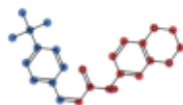
#0 Nc1ccc2c(c1)OCCO...



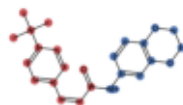
#1 Cc1ccc(C(C)(C)C)...



#2 CC(=O)Nc1ccc2c(c...



#3 CC(C)(C)c1ccc(/C... ●

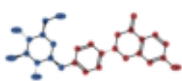
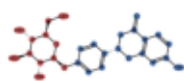


A1ETR · Q8NER1

Liquiritin

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 9VTZ
fold 0 sites 1 pocket 75.0 res
flavour (none) [llm_abstain]
SMILES O=C1C[C@@H](c2ccc(O[C@@H]3O[C@@H](CO)[C@@...

#0 OC[C@H]1O[C@@H](... #1 O=C1C[C@@H](c2cc...

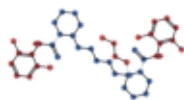


A1LZ2 · Q8NER1

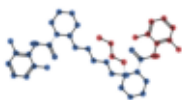
(1~{R},2~{S})-1-butyl-~{N}-(2,6-dimethylphenyl)-1-[5...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 8YCP
fold 0 sites 4 pocket 95.8 res
flavour (none) [llm_abstain]
SMILES CCCC[N@@+]1(CCCCN2CCCC[C@H]2C(=O)Nc2c(C...

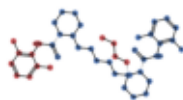
#0 CCCC...



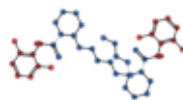
#1 CCCC...



#2 CCCC...



#3 Cc1cccc(C)c1N...



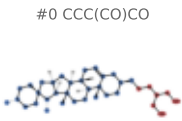
Catalogue

records 181–184 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

DUO · Q8NER1

2-[2-[(1~{S},2~{S},4~{S},5'~{R},6~{R},7~{S},8~{R},9~...

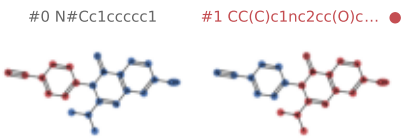
family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 8GFA
fold 0 sites 4 pocket 45.0 res
flavour (none) [llm_abstain]
SMILES C[C@@H]1CC[C@@]2(OC1)O[C@H]1C[C@H]3[C@@H...



EZI · Q8NER1

4-(7-Hydroxy-2-isopropyl-4-oxoquinazolin-3(4H)-yl)be...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 8JQR 8X94
fold 0 sites 8 pocket 68.8 res
flavour (none) [llm_abstain]
SMILES CC(C)c1nc2cc(0)ccc2c(=O)n1-c1ccc(C#N)cc1



LFA · Q8NER1

EICOSANE

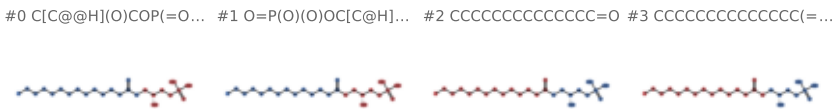
family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 11CL
fold 0 sites 4 pocket 54.0 res
flavour fatty [measured]
SMILES CCCCCCCCCCCCCCCC



NKN · O35433

(2R)-2-hydroxy-3-(phosphonoxy)propyl tetradecanoate

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 8T0C 8T0Y 8T10 8T3L 8T3M
fold 0 sites 12 pocket 89.2 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCCCC(=O)OC[C@H](O)COP(=O)(O)O



Catalogue

records 185–188 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XJ7 · O35433

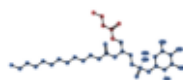
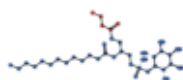
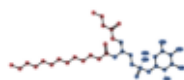
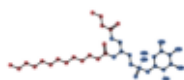
(2S)-1-(butanoyloxy)-3-[[R]-hydroxy{[(1r,2R,3S,4S,5...

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 7L2H 7L2J 7L2P 7L2S 7MZ6 7MZ9 7MZA 7MZB..
fold 0 sites 28 pocket 98.7 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCCCC(=O)O[C@@H](COC(=O)CCC)COP.

#0 CCCCCCCCCCCCCC=O... #1 CCCCCCCCCCCCCC=O...

#2 CCCC=O

#3 CCCC(=O)O



XJ7 · O35433;POCH43

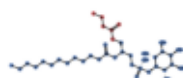
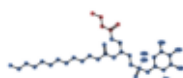
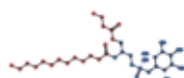
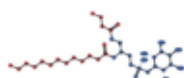
(2S)-1-(butanoyloxy)-3-[[R]-hydroxy{[(1r,2R,3S,4S,5...

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 7L2R 7L2T 7L2U
fold 0 sites 12 pocket 95.9 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCCCC(=O)O[C@@H](COC(=O)CCC)COP.

#0 CCCCCCCCCCCCCC=O... #1 CCCCCCCCCCCCCC=O...

#2 CCCC=O

#3 CCCC(=O)O



XJA · O35433

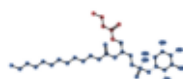
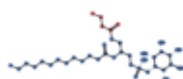
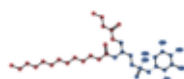
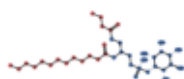
(2S)-1-(butanoyloxy)-3-[[R]-hydroxy{[(1S,2R,3R,4R,5...

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 7L2I
fold 0 sites 4 pocket 98.8 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCCCC(=O)O[C@@H](COC(=O)CCC)COP.

#0 CCCCCCCCCCCCCC=O... #1 CCCCCCCCCCCCCC=O...

#2 CCCC=O

#3 CCCC(=O)O



XJD · O35433

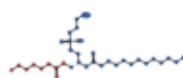
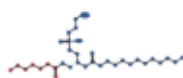
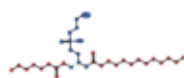
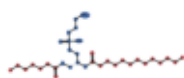
(10R,13S)-16-amino-13-hydroxy-7,13-dioxo-8,12,14-tri...

family TRPV1 — capsaicin/vanilloid receptor..
organism Rattus norvegicus
PDB 7L2H
fold 0 sites 4 pocket 52.0 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCCCCC(=O)O[C@H](COC(=O)CCCCC)CO.

#0 CCCCCCCCCCCCCC=O... #1 CCCCCCCCCCCCCC=O...

#2 CCCCCC=O

#3 CCCCCC(=O)O



Catalogue

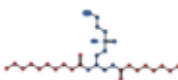
records 189–192 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XKP · O35433;P0CH43

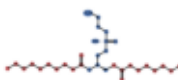
(11R,14S)-17-amino-14-hydroxy-8,14-dioxo-9,13,15-tri...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 7L2U
fold 0 sites 4 pocket 48.2 res
flavour (none) [llm_abstain]
SMILES CCCCCCCCC(=O)O[C@H](COC(=O)CCCCCC)C0[P..

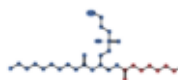
#0 CCCCCCCCCC=O...



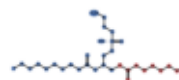
#1 CCCCCCCCCC=O...



#2 CCCCCCCC=O



#3 CCCCCCCC(=O)O

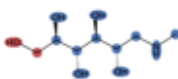


XPJ · O35433

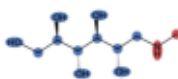
1-deoxy-1-(methylamino)-D-allitol

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 7L2X
fold 0 sites 1 pocket 53.0 res
flavour (none) [llm_abstain]
SMILES CNC[C@H](O)[C@H](O)[C@H](O)[C@H](O)CO

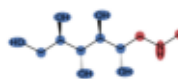
#0 CO



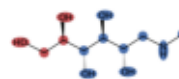
#1 CN



#2 CNC



#3 OCCO

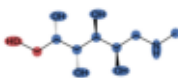


XPS · O35433

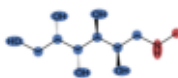
6-deoxy-6-(methylamino)-D-galactitol

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 7L2V
fold 0 sites 1 pocket 43.0 res
flavour (none) [llm_abstain]
SMILES CNC[C@@H](O)[C@H](O)[C@H](O)[C@@H](O)CO

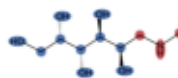
#0 CO



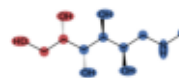
#1 CN



#2 CNC



#3 OCCO

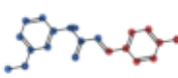


ZEI · Q8NER1

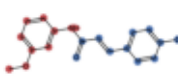
(2E)-3-(4-chlorophenyl)-N-(3-methoxyphenyl)prop-2-en...

family TRPV1 — capsaicin/vanilloid receptor...
organism Homo sapiens
PDB 8GFA
fold 0 sites 4 pocket 93.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc(NC(=O)/C=C/c2ccc(Cl)cc2)c1

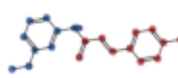
#0 Cc1ccc(Cl)cc1



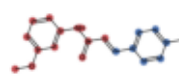
#1 COc1cccc(N)c1



#2 O=C/C=C/c1ccc(Cl)cc1



#3 CC(=O)Nc1cccc(OC)c1



Catalogue

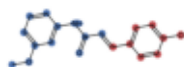
records 193–196 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

ZEI · O35433

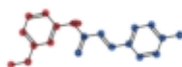
(2E)-3-(4-chlorophenyl)-N-(3-methoxyphenyl)prop-2-en...

family TRPV1 — capsaicin/vanilloid receptor...
organism Rattus norvegicus
PDB 9W4N
fold 0 sites 4 pocket 80.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc(NC(=O)/C=C/c2ccc(Cl)cc2)c1

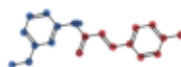
#0 Cc1ccc(Cl)cc1



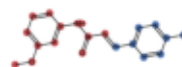
#1 COc1cccc(N)c1



#2 O=C/C=C/c1ccc(Cl)cc1



#3 CC(=O)Nc1cccc(OC)c1

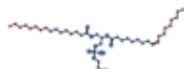


6OU · Q8NET8

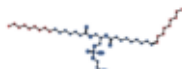
[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3...

family TRPV3 — warmth/camphor receptor...
organism Homo sapiens
PDB 6UW4 7XJ0 7XJ1 9BKU
fold 0 sites 86 pocket 43.8 res
flavour (none) [llm_abstain]
SMILES CCCCCCCC/C=C\CCCCCCCC(=O)O[C@H](COC(=O)C...

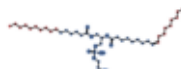
#0 CCCCCCCCC...



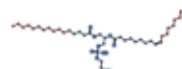
#1 CCCCCCCCC...



#2 CCCCCCCCC...



#3 CCCCCCCCC...

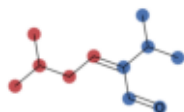


A1EBM · Q8NET8

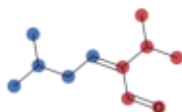
(~{Z})-5-methyl-2-propan-2-yl-hex-2-enal

family TRPV3 — warmth/camphor receptor...
organism Homo sapiens
PDB 9JEG
fold 0 sites 8 pocket 52.5 res
flavour floral;green;herbal;woody [measured]
SMILES CC(C)C/C=C\C(=O)C(C)C

#0 CCC(C)C



#1 CC(C)CC=O

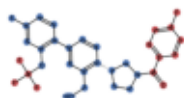


A1EYQ · Q8NET8

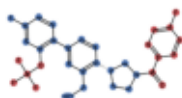
(5-fluoranylpyridin-2-yl)-[(3~{R})-3-[4-[4-fluoranyl...

family TRPV3 — warmth/camphor receptor...
organism Homo sapiens
PDB 9WV6 9WV7
fold 0 sites 8 pocket 83.0 res
flavour (none) [unlabelled_new]
SMILES O=C(c1ccc(F)cn1)N1CC[C@H](c2ccc(-c3ccc(F...

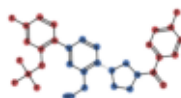
#0 O=Cc1ccc(F)cn1... ●



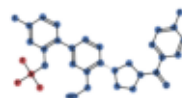
#1 O=Cc1ccc(F)cn1... ●



#2 O=Cc1ccc(F)cn1... ●



#3 FC(F)F



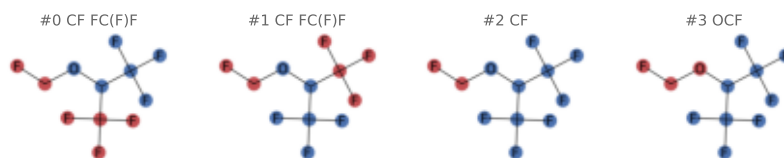
Catalogue

records 197-200 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1IPI · Q8NET8

sevoflurane

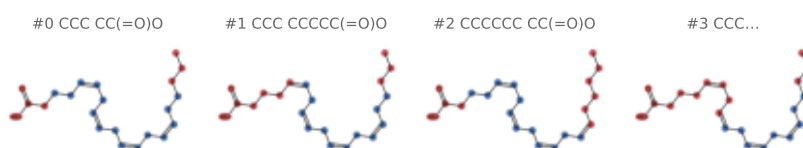
family	TRPV3 - warmth/camphor receptor...
organism	Homo sapiens
PDB	9UED
fold 0	sites 4 pocket 48.0 res
flavour	(none) [unlabelled_new]
SMILES	<chem>FCOC(C(F)(F)F)C(F)(F)F</chem>



ACD - Q8NET8

ARACHIDONIC ACID

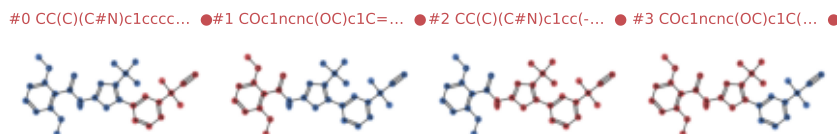
```
family      TRPV3 — warmth/camphor receptor...
organism    Homo sapiens
PDB         8V6N
fold 0      sites 4   pocket 63.0 res
flavour     (none) [llm_abstain]
SMILES      CCCCC/C=C\C/C=C\C/C=C\C/C=C\CCCC(=O)O
```



EQK · Q8NET8

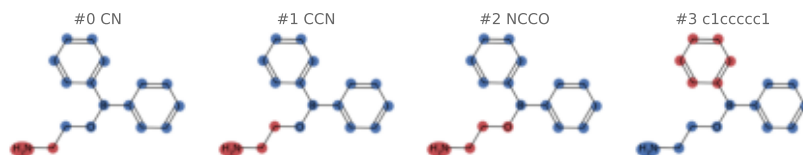
N-[5-[2-(2-cyanopropan-2-yl)pyridin-4-yl]-4-(trifluo...

family	TRPV3 — warmth/camphor receptor...
organism	Homo sapiens
PDB	7XJ0 7XJ1
fold 0	sites 14 pocket 79.8 res
flavour	(none) [unlabelled_new]
SMILES	CCc1ncnc(OC)c1C(=O)Nc1nc(C(F)(F)F)c(-c2c...

**FZ4 · Q8NET8**

2-aminoethyl diphenylborinate

```
family      TRPV3 - warmth/camphor receptor...
organism    Homo sapiens
PDB         8V6N 8V6O
fold 0      sites 16   pocket 48.2 res
flavour     unknown [measured]
SMILES      NCCOB(c1ccccc1)c1ccccc1
```



Catalogue

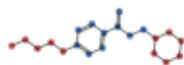
records 201–201 of 201 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

N8R · Q8K424

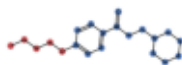
Dyclonine

family TRPV3 – warmth/camphor receptor...
organism Mus musculus
PDB 7UGG
fold 4 sites 4 pocket 59.2 res
flavour bitter [measured]
SMILES CCCC0c1ccc(C(=O)CCN2CCCCC2)cc1

#0 CCCCCO C1CCNCC1



#1 CCCCCO



#2 C1CCNCC1

